

**INTEGRATING ERC-20 ALLOWANCE EDGES INTO
PAGERANK FOR ENHANCED ON-CHAIN WALLET
REPUTATION SCORING**

by

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Abstract

Decentralized finance (DeFi) protocols increasingly rely on wallet-level reputation to gate integrations, calibrate risk parameters and screen for automated or adversarial behaviour. Existing on-chain reputation methods, exemplified by the transfer-graph Adaptive Weighted PageRank (AWP) of Do, Do and Nguyen (2023), aggregate historical token transfers. That design is costly to refresh on a rollup that emits millions of transfers per month, and transfer volume measures economic throughput rather than an explicit act of trust. The problem addressed in this thesis is whether a cheaper endorsement signal that already exists on every EVM chain, the ERC-20 allowance an owner grants to a spender, can support a reputation score that is operationally interpretable, computationally light and empirically checkable against public ledger data.

The thesis introduces EndorseRank, PageRank computed on a directed endorsement graph whose edges are the latest non-zero ERC-20 allowances between wallets. Following a quantitative, positivist design, EndorseRank is compared with a faithful re-implementation of AWP on the same matched cohort of $n = 5,521$ Arbitrum One wallets that have non-zero connectivity in both graphs over the window 1 December 2025 to 31 May 2026. Alignment with contemporaneous transfer, allowance and Sybil-stability statistics is measured with Kendall’s τ ; uncertainty is quantified with a paired wallet bootstrap (400 resamples) for every coefficient and for six pre-specified differences between coefficients. A temporal holdout scores wallets on events up to 28 February 2026 and tests the scores against owner–spender approvals that first appear in March–May 2026. Runtime is measured under a single timing protocol (one warm-up, five timed solves), and node-count sensitivity is examined on an expanded pool of $N = 99,080$ addresses.

Three results follow. First, EndorseRank solves PageRank in 1.43 s against 11.25 s for AWP on the same cohort (about 7.9 times faster), because its graph has 14,727 edges against 137,087; the ratio stays between 7.9 and 10.5 across staged node counts. Second, the two methods measure different

constructs: AWP aligns more strongly with transfer proxies ($\tau = 0.534$ against 0.333 for EndorseRank), whereas EndorseRank aligns with allowance proxies ($\tau = 0.355$, 95% CI 0.322–0.384) where AWP is slightly negative (-0.030); the inter-method agreement is $\tau = 0.316$. EndorseRank’s allowance alignment is not statistically higher than its own transfer alignment, so the allowance result is read as an intended-construct check, not as an external test. Third, on the holdout spender cohort ($n = 1,335$) EndorseRank ranks wallets that later receive new approvals at $\tau = 0.479$ (95% CI 0.426–0.529) while AWP is near zero (0.044). Damping, top-20-token and sample-definition checks do not reverse these patterns.

The comparison is the first same-cohort evaluation of an allowance-based and a transfer-based PageRank that reports confidence intervals, difference tests, a temporal holdout and a versioned pipeline. A reputation score can be refreshed at roughly one eighth of the cost of the transfer-graph baseline and still capture a trust act that the baseline misses. Cheaper, more frequently refreshed authorization-aware scores lower the barrier to risk screening that today favours well-resourced actors. For protocol operators and regulators, an auditable score built on public approvals is a transparent complement to opaque proprietary risk engines. EndorseRank does not outperform AWP on transfer-hub detection. Its Sybil resistance depends on the economic cost of manufacturing approvals, which is modelled but not measured here.

Keywords— DeFi, EndorseRank, Adaptive Weighted PageRank, on-chain reputation, ERC-20 allowance, temporal holdout, Arbitrum One, Kendall’s tau

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Abbreviations and Notation

This section lists abbreviations and mathematical notation used throughout this research. Symbols are defined again at first use in the main chapters when helpful for local reading.

Abbreviations

Table 1: Abbreviations used in this research.

Abbreviation	Meaning
AMM	Automated Market Maker
AWP	Adaptive Weighted PageRank
BQ	BigQuery
CCS	ACM Conference on Computer and Communications Security
DeFi	Decentralized Finance
DEX	Decentralized Exchange
EIP	Ethereum Improvement Proposal
ERC-20	Ethereum Request for Comments 20 (fungible token standard)
FC	Financial Cryptography and Data Security
LP	Liquidity Provider
MEV	Maximal Extractable Value (formerly Miner Extractable Value)
PR	PageRank
RQ	Research Question
SoK	Systematization of Knowledge
SRQ	Supplementary Research Question
USENIX	USENIX Security Symposium
WWW	World Wide Web Conference

Mathematical notation

Table 2: Mathematical notation used in this research.

Symbol	Meaning
G_{approve}	Directed endorsement graph from latest ERC-20 allowances (owner $\rightarrow$ spender)
G_{transfer}	Directed transfer graph with time-decayed edge weights (AWP input)
n	Alignment-cohort size (primary comparison: $n = 5,521$)
N	Expanded wallet-pool size (scaling: $N = 99,080$)
d	PageRank damping (teleportation) factor
k	AWP logistic time-decay steepness
t_0	AWP logistic time-decay midpoint (days)
Δt	Edge age in days for AWP decay
$\sigma(\Delta t)$	Logistic decay weight applied to transfer edges
ρ	Spearman’s rank correlation coefficient
τ	Kendall’s rank correlation coefficient
$ E $	Number of directed edges in a reputation graph
$\mathcal{W}$	Set of wallets in the evaluation cohort
$s(w)$	Reputation score of wallet w under a given method
$p(w)$	Validation-proxy value for wallet w

1 Introduction and Research Problem

Decentralized Finance (DeFi) changed how capital is allocated on public blockchains: lending, borrowing, and leveraged trading can run without traditional financial intermediaries ^[1]. Yet most credit-like interactions remain highly collateralized because protocols cannot rely on off-chain identity systems or conventional credit histories ^[2]. In a pseudonymous environment, the operational challenge is to allocate capital safely when counterparties are addresses rather than identified persons.

This research addresses a concrete research problem within that setting: whether an on-chain reputation score built from ERC-20 approve/permit allowances—implemented as EndorseRank—can reduce computational cost, represent endorsement semantics more faithfully, and yield a distinct yet credible alignment profile compared with transfer-based Adaptive Weighted PageRank (AWP) on Arbitrum One. The primary empirical comparison is EndorseRank versus AWP on an identical wallet sample, checked with contemporaneous transfer and allowance statistics and with a temporal holdout of future new approvals, under the rank-correlation validation protocol introduced with AWP^[3].

In this study, on-chain reputation score computation refers to deriving a quantitative wallet score from observable on-chain behavior and relationships without off-chain identity ^[4]. EndorseRank produces such a score from allowance-based endorsement edges. The remainder of this chapter situates the problem historically and economically, motivates allowance-based reputation, states the research problem, objectives, and questions, defines key constructs, summarizes the literature and the methodology that later chapters develop, and outlines contributions, scope, and the research roadmap.

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains introduced programmable settlement and open participation without a central operator^[5]. Smart contracts encode agreement terms as self-executing code^[6], enabling financial applications that settle on-chain rather than through custodial intermediaries^[7]. DeFi protocols compose these contracts into markets for lending, automated market making, and perpetual futures^[8]. On Arbitrum One and similar Layer-2 networks, activity density is high enough that wallet-level behavioral signals—transfers, approvals, and protocol events—can be observed at scale.

Despite this transparency, DeFi credit markets remain structurally over-collateralized. Protocols typically require borrowers or traders to post collateral exceeding the value of their exposure because counterparties are pseudonymous addresses rather than identified legal persons^{[9],[10]}. Pseudonymity is a design feature of public ledgers: anyone may create addresses at negligible cost, and a single economic actor may control many wallets^[11]. Traditional credit scoring depends on identity-linked histories that do not transfer cleanly to this setting^[2]. Consequently, risk management defaults to collateral buffers, liquidation engines, and protocol-level parameters rather than individualized reputation.

Collateralization protects protocols but imposes capital inefficiency. Traders and borrowers must lock assets that could otherwise be deployed elsewhere; protocols forgo under-collateralized products that traditional finance offers to borrowers with established credit. Researchers have therefore explored on-chain signals as substitutes for off-chain identity: transfer-graph centrality^[4], network risk propagation^[12], prospect-theory and blockchain-based credit models^[13], and machine-learning pipelines on lending protocols^[14]. These approaches share a premise: public ledger data encode economically meaningful relationships that can inform risk assessment without deanonymizing users.

Graph ranking methods are a natural fit for this premise. PageRank and its variants propagate influence through directed edges, ranking nodes by recursive endorsement from other nodes^[15]. Early web-trust systems such as EigenTrust^[16] and TrustRank^[17] applied similar ideas to peer-to-

peer and web spam settings. Do and Do^[4] apply PageRank and HodgeRank to Ethereum transfer graphs as a social-credit measure. Adaptive Weighted PageRank (AWP), proposed by Do, Do, and Nguyen^[3], is a separate transfer-graph ranking method that uses value-weighted, logistically time-decayed edges and validates scores against inbound transfer count and value as domain-intuition proxies for economic importance. That line of work establishes transfer-graph reputation as a credible baseline. It does not model explicit spending authorization, a distinct on-chain act recorded by ERC-20 allowances ^{[18],[19]}.

1.2 Motivation for on-chain reputation

Four practical pressures motivate on-chain reputation research in leveraged DeFi.

- Capital efficiency: protocols seek ordinal signals that could support differentiated service—for example, limit-increase cadence, fee tiers, or monitoring priority—and, only under future controlled experiments, modest collateral differentiation, while preserving on-chain settlement ^[1].
- Auditability: unlike opaque off-chain credit scores, graph-based scores derived from public events can be recomputed and inspected, addressing concerns about black-box decentralized credit scoring ^[2].
- Operational cost: reputation systems that must refresh frequently—for example, before each leverage adjustment—cannot afford expensive historical transfer aggregation on every update.
- Standardization: pseudonymous DeFi lacks a shared ordinal vocabulary for wallet-level counterparty differentiation of the kind that screening ratios provide in other markets. Chapter 5 develops industry adoption scenarios in which EndorseRank and AWP play that infrastructural role—supporting fees, utilization, and monitoring efficiency—without substituting for collateral.

Transfer-based AWP addresses the first two pressures but is costly on the third. Constructing G_{transfer}

requires aggregating transfer history, applying logistic time-decay, and running PageRank on a dense edge set ^[3]. Allowance graphs offer a structurally different alternative. An ERC-20 `approve` or EIP-2612 `permit` records how much a token owner authorizes a spender to transfer ^{[18],[19]}. The latest allowance per owner–spender pair is a snapshot of current authorization intent: newer approvals override older ones, so temporal decay is unnecessary. Because approvals are rarer on-chain events than transfers, the resulting endorsement graph G_{approve} has roughly an order of magnitude fewer edges than G_{transfer} on the same wallets, and PageRank—whose per-iteration cost is $O(|E|)$ —completes correspondingly faster (Table 4.1). That gap is an expected consequence of graph sparsity under latest-allowance snapshots, not an algorithmic contribution: the solver is identical for both methods. Claim C1 reports it as an operational property of the representation. Beyond efficiency, allowances carry distinct semantics. A transfer may reflect routing, market making, or passive receipt; an allowance is an explicit delegation of spending authority. Treating allowances as endorsement edges aligns PageRank’s citation metaphor ^[15] with a trust-like act that protocols already rely on for token interactions. Whether that semantic difference produces rankings that differ, and whether those rankings align differently with authorization and transfer checks, is an empirical question. This research answers it on Arbitrum One using a matched sample of wallets with non-zero endorsement- and transfer-graph connectivity ($n = 5,521$).¹

1.3 Research problem

Existing Adaptive Weighted PageRank (AWP) methods infer trust from transfer volume and logistic time-decay over transaction history ^[3]. While economically interpretable, they (i) require expensive historical transfer aggregation, (ii) treat inbound transfer volume as a proxy for a “valuable wallet,” and (iii) do not model explicit spending authorization. EndorseRank instead uses the latest allowance per owner–spender pair as edge strength on a directed endorsement graph, yielding a sparser graph and a semantically distinct trust signal without temporal decay weighting.

¹The matched cohort is wallets with non-zero connectivity in the endorsement and transfer subgraphs in the observation window. Runtime sensitivity to node count uses a larger expanded wallet pool ($N = 99,080$).

The research problem is therefore comparative and evaluative: on an identical Arbitrum One wallet sample, does EndorseRank produce wallet orderings that are construct-distinct from AWP, align with allowance and transfer checks, predict later new endorsements in a time holdout, and do so at substantially lower computational cost? The primary comparison is EndorseRank versus AWP. Table 1.1 summarizes the primary comparison. Table 1.2 summarizes hypotheses (H1–H3), research questions (RQ1–RQ4), and Chapter 4 empirical claims (C1–C4).

Table 1.1: Primary comparison: EndorseRank versus AWP (matched wallet cohort, $n = 5,521$).

Method	Input graph	Role
EndorseRank	Latest ERC-20 allowances (G_{approve})	Primary construct: allowance delegation; target of this research
AWP	Time-decayed transfers (G_{transfer})	Primary baseline: IEEE RIVF 2023 transfer-graph method [3]; Do and Do ^[4] is the earlier PageRank/HodgeRank study on Ethereum transfers

To avoid ambiguity, the key concepts used throughout this research are defined as follows:

- *DeFi (Decentralized Finance)*: Financial services—including lending and borrowing—implemented via smart contracts on public blockchains, typically without a central operator controlling custody or settlement [1]. DeFi protocols compose open interfaces so that token movements are observable on-chain.
- *Smart contract*: Self-executing program code deployed on a blockchain that enforces agreement terms without a trusted intermediary [6]. In this research, smart contracts are the source of ERC-20 allowance logs and transfer logs.
- *Wallet / Address*: A blockchain account that holds assets and interacts with smart contracts; the unit of analysis for reputation scoring [7]. A wallet is identified by its address; no off-chain identity is assumed or recovered.
- *On-chain reputation score*: A numerical indicator derived from on-chain signals that represents a wallet’s relative standing for financial interactions under pseudonymity [4]. Scores are ordinal for ranking purposes; absolute magnitudes are not interpreted as credit limits.

- *PageRank*: A graph ranking method that propagates influence through directed edges under a damping factor^[15]. EndorseRank applies PageRank to a latest-allowance endorsement graph: edge strength equals the current allowance per owner–spender pair, with no temporal decay. Adaptive Weighted PageRank (AWP) applies PageRank to transfer graphs with edge weights given by transfer value and logistic time-decay^[3].
- *ERC-20 allowance / Approve / Permit*: Mechanisms recording how much a token owner permits a spender to transfer on the owner’s behalf^{[18],[19]}. Approve is a standard on-chain call; EIP-2612 permit authorizes the same state change via signature. EndorseRank uses the latest non-zero allowance snapshot per owner–spender pair.
- *Endorsement graph*: A directed graph in which edges represent explicit delegation of spending authority rather than mere asset transfers^[18]. In EndorseRank, an edge $u \rightarrow v$ means wallet u currently authorizes wallet or contract v to spend tokens owned by u ; transfer-graph AWP provides the contrasting baseline^[3].
- *Domain alignment*: The degree to which a reputation ranking agrees with an observable proxy ranking, assessed by Spearman’s ρ ^[20] and Kendall’s τ ^[21], in the construct-validity sense of Cronbach and Meehl^[22]. Alignment is construct-specific: a method may align strongly with its primary graph construct and weakly with an orthogonal check.

1.4 Temporal holdout

A contemporaneous allowance Kendall τ (Table 4.6) is an intended-construct check: PageRank on G_{approve} is expected to agree with spender-side approval degree because the score is a global smoothing of that degree. To test whether the score carries information beyond the same-window graph, scores are frozen at 28 February 2026 and compared with new owner–spender approval pairs that first appear in March–May 2026. The claimed holdout cohort is t1 inbound spenders ($n = 1,335$).

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1 / RQ1	Do EndorseRank and AWP yield distinct wallet orderings?	§4.5; inter-method τ with 95% CI
H2 / RQ2	Which social method aligns with allowance vs. transfer checks?	Tables 4.3 and 4.4; Claims C2–C3
H3 / RQ3	Is EndorseRank faster than AWP at equal n ?	Tables 4.1–4.2; Claims C1, C4
RQ4	Do t1 scores predict t2 new approvals beyond same-window allowance?	§4.7; Table 4.11
C1	EndorseRank vs. AWP runtime and graph size	Table 4.1
C2	Intended-construct checks: allowance for ER, transfer for AWP	Tables 4.5–4.6
C3	Construct-specific winners (EndorseRank vs. AWP)	Table 4.3
C4	Efficiency–alignment trade-off for EndorseRank	Tables 4.1, 4.3 and 4.11

1.5 Research objectives

The study has four primary objectives, mapped one-to-one onto RQ1–RQ4. Each objective names the measurement that operationalizes it and the criterion by which Chapter 5 judges whether it has been met.

- O1. Determine whether EndorseRank and AWP induce distinct wallet orderings on an identical Arbitrum One cohort (RQ1). Measurement: Kendall τ between the two score vectors on the matched cohort ($n = 5,521$) with a 95% bootstrap confidence interval. Criterion: the interval excludes both 0 (unrelated orderings) and 1 (identical orderings).
- O2. Measure contemporaneous construct checks—allowance for EndorseRank and transfer for AWP—on the same cohort (RQ2). Measurement: family-level Kendall τ with 95% intervals and paired bootstrap contrasts $\Delta\tau$ between methods within each family. Criterion: each method’s $\Delta\tau$ on its own construct is positive with an interval excluding 0.
- O3. Compare PageRank computational cost of EndorseRank and AWP at equal n , including node-count sensitivity on the expanded pool (RQ3). Measurement: wall-clock solve time (mean

and SD over five timed repeats after one warm-up), peak memory, iterations, and $|V|$, $|E|$ per configuration under one fixed protocol. Criterion: the runtime ratio is reported together with the edge ratio so that any speed advantage is attributed to sparsity.

- O4. Test whether t1 scores predict t2 new owner–spender approvals on the spender cohort (RQ4). Measurement: Kendall τ between scores frozen at 28 February 2026 and new-approval counts in March–May 2026 on $n = 1,335$ inbound spenders, with a 95% interval, compared against AWP and against the same-window allowance degree. Criterion: EndorseRank’s interval excludes 0 and its point estimate exceeds the AWP estimate.

1.6 Research questions

This study tests three primary hypotheses: H1—EndorseRank and AWP yield distinct wallet orderings on the same cohort; H2—the two social methods differ in construct-specific checks (allowance versus transfer); and H3—EndorseRank is computationally more efficient than AWP at equal n . The following research questions guide the empirical evaluation summarized in Table 1.2.

- RQ1. Do EndorseRank and AWP produce meaningfully different wallet orderings on an identical Arbitrum wallet sample?
- RQ2. Between EndorseRank and AWP, which social method aligns more strongly with contemporaneous allowance checks and with contemporaneous transfer checks (Spearman’s ρ , Kendall’s τ)?
- RQ3. Is EndorseRank computationally more efficient than AWP on the same matched sample (runtime, memory, iterations)?
- RQ4. After scores are frozen at t1, does EndorseRank predict t2 new owner–spender approvals on inbound spenders more strongly than AWP, and more informatively than the same-window allowance check?

RQ1 tests whether allowance and transfer graphs induce distinct centrality structures, operationalized by inter-method Kendall τ between score vectors. RQ2 evaluates intended-construct checks

on transfer and allowance local statistics; because each method’s own-construct proxies are derived from the same event stream that builds its graph, RQ2 is read as a construct-validity observation, not as external validation. RQ3 compares PageRank wall-clock time, memory, and iterations on identical wallet sets, including node-count sensitivity on an expanded wallet pool. RQ4 is a time-split test of the same endorsement construct: future new approvals, not trading profit. RQ4 replaces the trading-success comparison of the research proposal, which was withdrawn because the outcome-native reference it relied on was circular with the proxies (Section 2.8).

1.7 Summary of the literature review

Chapter 2 reviews four bodies of work; the following summary states what each contributes to the design and where it falls short. First, graph ranking. PageRank^[15] ranks nodes by recursively propagated endorsement and is computed by power iteration whose per-iteration cost is linear in the number of edges^[23]; EigenTrust^[16] and TrustRank^[17] adapt the idea to peer-to-peer trust and web spam with normalisation and seed sets that limit inflation. These methods supply the solver and the complexity argument used here, but their edge semantics are fixed to hyperlinks, ratings, or seeds. Second, on-chain reputation. Do and Do^[4] apply PageRank and HodgeRank to Ethereum transaction graphs as a social-credit measure; Do, Do, and Nguyen^[3] propose AWP, which weights transfer edges by value and a logistic decay of age and validates the ranking against inbound transfer count and value; Lin et al.^[12] propagate account risk through the transaction network; Wu et al.^[24] show that graph embeddings of transaction data outperform per-account features for phishing detection. Surveys of blockchain reputation systems^{[25],[26]} confirm that explicit ratings and transfer volume are the dominant trust signals. None of these works uses the ERC-20 allowance—the on-chain act that records a deliberate delegation of spending authority^[18]—as the graph edge, and the AWP validation compares scores with proxies computed from the same transfer data that builds the graph. Third, DeFi risk and measurement. Systematizations of DeFi^[8] and of DeFi attacks^[27] locate behavioral counterparty risk as one layer among smart-contract, oracle, market, and

governance risk; wash trading ^{[28],[29]}, extractable value ^[30], scam tokens ^[31], and airdrop farming ^[32] show how much transfer activity is artificial, which is the empirical reason to look for a signal other than payment flow. Fourth, measurement theory and inference. Construct validity ^{[22],[33]} distinguishes what a score is designed to measure from external criteria, and bootstrap resampling ^{[34],[35]} provides distribution-free intervals for rank statistics. The gap that emerges (Section 2.21) is a same-cohort comparison of an allowance-edge ranking against the transfer-edge baseline, with confidence intervals, paired difference tests, a fixed timing protocol that reports edge counts, and a temporal holdout that separates same-window agreement from prediction.

1.8 Summary of the research methodology

Chapter 3 adopts a quantitative, comparative design on public ledger data. ERC-20 Approval and Transfer logs for Arbitrum One over 1 December 2025–31 May 2026 are extracted from Google BigQuery; the matched wallet cohort ($n = 5,521$) comprises addresses with non-zero connectivity in both the latest-allowance graph G_{approve} and the time-decayed transfer graph G_{transfer} . Both graphs are scored with one shared power-iteration PageRank implementation ($d = 0.85$, $\varepsilon = 10^{-8}$, at most 300 iterations); EndorseRank uses summed latest allowances as edge weights, AWP uses value-weighted, logistically decayed transfers exactly as specified in the baseline paper^[3]. Rank agreement is measured with Spearman’s ρ and Kendall’s τ against three contemporaneous proxy families (transfer, allowance, Sybil-adjusted stability) and, for RQ4, against new owner–spender approvals in a three-month holdout after scores are frozen. Every coefficient carries a 95% percentile bootstrap interval from 400 paired wallet resamples (seed 42), and pre-specified method contrasts are reported as paired $\Delta\tau$ intervals. Runtime, peak memory, iterations, and $|V|$, $|E|$ are recorded under one timing protocol (one untimed warm-up, five timed repeats, per-session memoization so that identical graphs yield identical numbers wherever they are tabulated). Sensitivity analyses vary damping, restrict to top tokens, and change the wallet-sample definition. The pipeline, configuration, and machine-readable results summary from which every table is generated are ver-

sioned in a public repository (Section 3.13); the study uses public data only and involves no human participants (Section 3.16).

1.9 Contributions

This research makes three contributions to on-chain reputation research and DeFi risk assessment. EndorseRank applies classical PageRank^[15] unchanged. The contribution is the combination of a new edge construct, a same-cohort comparison with inference, and an out-of-window test.

- Latest-allowance edge construction (EndorseRank): an endorsement graph in which the edge $u \rightarrow v$ carries the current ERC-20 approve/permit allowance from owner u to spender v . Newer approvals override older ones, so the construct needs no temporal decay, and it records explicit delegation of spending authority rather than payment flow. No reviewed reputation method uses this edge (Section 2.21).
- Same-cohort contemporaneous comparison with inference: EndorseRank and AWP are computed on one matched wallet cohort and compared on identical transfer, allowance, and Sybil-stability proxies under the AWP validation protocol^[3], with 95% bootstrap intervals on every coefficient and paired $\Delta\tau$ contrasts^[34]. The design states that each method’s own-construct agreement is partly mechanical, and it reports edge counts alongside runtimes so that the efficiency result (Claim C1) is read as an expected consequence of sparsity.
- Temporal holdout of future new approvals: scores frozen at t_1 are tested against new owner–spender approvals in t_2 on the spender cohort, separating same-window agreement from prediction. Together with the open, versioned evaluation pipeline (Section 3.13; Appendix 8.1), this provides the evidence base for Claims C1–C4.

Headline numerical results for inter-method τ , family-level alignment, and runtime are reported once in Chapter 4 (Tables 4.1 and 4.3) and are not restated here. Those results describe an efficiency–alignment trade-off: neither social method dominates both constructs.

1.10 Scope and limitations

This research focuses on Arbitrum One over the observation window 2025-12-01 to 2026-05-31. The primary empirical sample is a matched wallet cohort of $n = 5,521$ addresses with non-zero endorsement- and transfer-graph connectivity; edge counts for both graphs are reported in Table 4.1. Runtime sensitivity to node count is reported on an expanded wallet pool ($N = 99,080$). Sensitivity analyses examine damping, top-token subgraphs, and the wallet-sample definition.

Unsecured lending default labels are out of scope. Constructs and ethics are defined in Chapters 1 and 3. The holdout claim uses the spender cohort ($n = 1,335$). Extended baselines archived for future comparison are excluded from the main narrative (Appendix 8.1).

Generalization beyond Arbitrum One and the six-month window is not claimed. Token-decimal normalization, price oracles, and permit-versus-approve coverage may affect edge weights. Adversarial manipulation of allowances or transfers remains a threat discussed in Chapter 5; Sybil-adjusted proxies partially address but do not eliminate that risk ^[11].

1.11 Research roadmap

This research proceeds as follows.

Chapter 2 reviews graph-based on-chain reputation and ERC-20 allowance semantics. It develops the theoretical framework—hyperlink propagation, domain alignment, and computational complexity—that motivates the primary EndorseRank versus AWP comparison.

Chapter 3 specifies data sources, sampling, graph construction for EndorseRank and AWP, contemporaneous construct checks, the temporal holdout, computational benchmarks, robustness checks, reproducibility practices, and ethics. It defines the matched wallet cohort, the spender holdout cohort, the expanded wallet pool used for scaling, and the operational procedures that produce the Chapter 4 tables.

Chapter 4 reports empirical results: cohort characteristics, runtime and scaling benchmarks (Claim C1), robustness checks, allowance and transfer alignment (Claims C2–C3), rank diver-

gence (H1/RQ1), and the temporal holdout (RQ4). Primary claims C1–C4 are stated explicitly.

Chapter 5 interprets findings by research question, develops theoretical and practical implications, formulates a Sybil attack cost model on allowance graphs, discusses regulatory and ethical considerations, and evaluates threats to validity and limitations.

Chapter 6 synthesizes the contribution to knowledge, answers RQ1–RQ4 in turn, and outlines future work on cross-protocol validation, adversarial robustness, hybrid methods, and live oracle integration.

Chapter 2 starts from the blockchain and DeFi primitives that the allowance construct uses and closes with the research gap this study is designed to close.

2 Literature Review and Theoretical Framework

Chapter 1 stated the research problem—that transfer-graph reputation scores conflate payment flow with trust—and derived objectives O1–O4 and research questions RQ1–RQ4 from it. This chapter reviews the conceptual and technical foundations required to interpret EndorseRank and Adaptive Weighted PageRank (AWP) and to judge whether those objectives are attainable with public data. It begins with blockchain and DeFi primitives for readers who may not specialize in distributed ledgers, then develops graph ranking theory, reputation-system lineage, ERC-20 allowance semantics, and rank-correlation statistics. A theoretical framework—hyperlink–citation propagation, domain alignment, and computational complexity—motivates the primary EndorseRank versus AWP comparison. Later sections supply a terminology guide, worked example, risk taxonomy, AWP in depth, a tabular summary of the reviewed literature (methodology, strengths, limitations), and an explicit statement of the research gap that Chapters 3 and 4 address.

2.1 Blockchain and DeFi foundations

A blockchain is a replicated append-only ledger in which batches of transactions (blocks) are linked by cryptographic hashes ^[5]. Participants agree on a canonical history through a consensus mechanism; once confirmed, entries are costly to rewrite. This property enables *trust-minimized* settlement: counterparties need not share a legal identity if they can verify the ledger state.

Bitcoin demonstrated peer-to-peer electronic cash ^[5]. Ethereum generalized the model to a world computer: accounts hold balances and code, and transactions invoke functions that update state ^[7]; the execution semantics are specified by the Ethereum Virtual Machine (EVM) ^[36]. Smart contracts are programs stored on-chain that execute deterministically when called ^[6]. Because execution is public and reproducible, financial logic—collateral checks, liquidations, fee accrual—can be

encoded without a central operator ^[1].

Decentralized Finance (DeFi) refers to the ecosystem of such contracts offering exchange, lending, derivatives, and asset management; Werner et al.^[8] give a systematization of the protocol classes, and Schär^[1] describes the layered architecture. Composability—the ability of one protocol to call another—creates powerful products ^[37] but also systemic risk pathways: flash-loan attacks ^[38] and price-oracle manipulation ^[39] are the canonical examples. For reputation research, the relevant implication is that *wallet-level histories* on public chains are rich, machine-readable, and incomplete: they reveal actions, not legal identities ^[2].

2.1.1 Pseudonymity and the credit problem

Traditional credit scoring links legal persons to repayment histories. On public blockchains, the unit of analysis is an address (wallet). An address may represent an individual, a fund, a bot, or a smart contract. Multiple addresses may be controlled by one entity ^[11], and clustering heuristics recover such control only partially ^[40]. Consequently, DeFi lending markets typically require over-collateralization: borrowers lock assets worth more than the loan, and liquidators seize collateral when health factors fall ^[41]. Over-collateralization is robust but capital-inefficient. On-chain reputation scores aim to summarize behavioral reliability so that protocols—or researchers studying them—can reason about risk without off-chain KYC; proposals range from prospect-theory credit models ^[13] to machine-learning scoring on lending data ^[14] and risk propagation over account networks ^[12].

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Ethereum mainnet provides security and liquidity but faces throughput and fee constraints under demand ^[42]. Layer-2 (L2) protocols execute transactions off the base layer (or in a compressed form) and periodically post commitments to Ethereum, aiming to inherit security while reducing cost ^[43]; Thibault et al.^[44] survey the rollup design space. Optimistic rollups such as Arbitrum

assume correctness unless challenged during a dispute window ^[45].

Arbitrum One is an EVM-compatible optimistic rollup widely used for DeFi activity, including the GMX perpetual-swap venue ^[46] whose contracts appear among the most-approved spenders in the data; perpetual futures themselves are a distinct derivative class whose funding-rate pricing is analysed by Ackerer et al.^[47]. For this research, Arbitrum One is the empirical setting because (i) ERC-20 Approval and Transfer logs are available in public BigQuery datasets and (ii) fee levels permit dense activity within a six-month window. Results should not be assumed to transfer automatically to other L2s or to Ethereum mainnet without re-validation, but the methodological template—allowance graph versus transfer graph, proxy families, rank correlation—is chain-agnostic wherever equivalent logs exist.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens: `transfer`, `approve`, `transferFrom`, `allowance`, and related events ^[18]. When an owner calls `approve(spender, amount)`, the token contract records that `spender` may transfer up to `amount` from the owner. The `spender` later calls `transferFrom` to move tokens. EIP-2612 permit allows allowance updates via off-chain signatures settled in a single transaction ^[19], improving user experience without changing the fundamental authorization semantics.

Figure 2.1 illustrates the lifecycle. An allowance is an *explicit authorization*, not a completed transfer. Owners may set allowances to routers, vaults, or trading contracts they trust to act within limits. Revocation (approving zero) and replacement (a new non-zero allowance) update the live authorization state. EndorseRank therefore uses the *latest* allowance per owner–spender pair as edge strength: older approvals are superseded rather than decayed continuously.

From a measurement perspective, allowances are sparser than transfers: many transfers occur under a single approval, and many wallets never approve third-party spenders. This sparsity is both a computational advantage and a semantic claim—EndorseRank measures authorization networks,

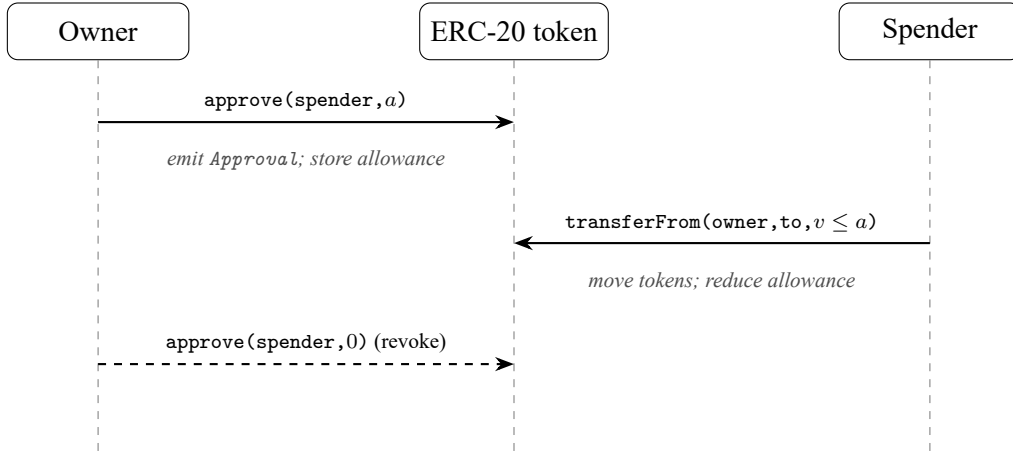


Figure 2.1: ERC-20 allowance lifecycle: approve (or permit), optional `transferFrom`, and revoke. EndorseRank edges reflect the latest live allowance snapshot per owner–spender pair (historical approvals superseded, not time-decayed).

not payment networks.

2.4 Graph-based on-chain reputation

PageRank and variants rank nodes by propagated influence through directed edges^[15]. Two recent lines of work apply this idea to blockchain transaction graphs and must be kept distinct. Do and Do^[4] apply PageRank and HodgeRank to Ethereum transaction graphs as a social-credit measure; their contribution is the framing of transfer connectivity as credit-relevant. Adaptive Weighted PageRank (AWP), proposed by Do, Do, and Nguyen^[3], is the transfer-graph baseline used in this thesis: it weights each transfer edge by value and by a logistic time-decay of edge age, and it validates the resulting scores against in-degree and in-value—count and sum of inbound transfers—as domain-intuition proxies for economic importance. The logistic decay formulation and the in-degree/in-value validation protocol reproduced in Chapter 3 are therefore attributed to the AWP paper^[3], not to Do and Do^[4]. Lin et al.^[12] show that transaction connectivity alone can miss risk-relevant structure unless risk signals propagate through the network, motivating refined graph semantics beyond vanilla PageRank; Wu et al.^[24] reach a similar conclusion for phishing detection, where network embeddings of transaction graphs outperform per-account features. Surveys of

blockchain-based reputation systems ^[25] and of smart-contract reputation designs ^[26] document that most deployed systems still rely on explicit ratings or transfer volume as the trust signal; a survey of knowledge discovery in cryptocurrency transaction data ^[48] likewise lists degree- and value-based centralities as the dominant wallet descriptors. Packin and Lev-Aretz^[2] emphasize auditability constraints under pseudonymity. EndorseRank stays within interpretable graph propagation. It is not a black-box predictive score.

Graph-based approaches are attractive because they (i) use only public data, (ii) produce continuous scores suitable for ranking, and (iii) admit complexity analysis in terms of edge counts. They are limited by Sybil attacks ^[11], wash trading, and the gap between graph centrality and economic default risk. PageRank scores are not treated as credit scores in the regulatory sense. The claim is that allowance-based and transfer-based rankings can be compared rigorously on shared proxies.

2.5 PageRank formalism

Let $G = (V, E)$ be a directed graph with $|V| = n$ nodes. Write $w_{ij} \geq 0$ for the weight of edge $i \rightarrow j$ (influence from i toward j in the endorsement or transfer sense used here). Let W be the weighted adjacency matrix and define the out-strength $d_i^{\text{out}} = \sum_j w_{ij}$. The row-stochastic transition matrix P is

$$P_{ij} = \begin{cases} w_{ij}/d_i^{\text{out}} & \text{if } d_i^{\text{out}} > 0, \\ 1/n & \text{if } d_i^{\text{out}} = 0 \end{cases} \quad (2.1)$$

(dangling nodes distribute mass uniformly). The Google matrix with damping $d \in (0, 1)$ is

$$G_{\text{PR}} = d P^\top + \frac{1-d}{n} \mathbf{1}\mathbf{1}^\top, \quad (2.2)$$

and the PageRank vector $\boldsymbol{\pi}$ satisfies $\boldsymbol{\pi} = G_{\text{PR}} \boldsymbol{\pi}$ with $\boldsymbol{\pi} \geq 0$ and $\sum_i \pi_i = 1$ ^{[15],[23]}. Equivalently,

$$\pi_j = d \sum_{i:(i \rightarrow j) \in E} \pi_i \frac{w_{ij}}{d_i^{\text{out}}} + \frac{1-d}{n}. \quad (2.3)$$

2.5.1 Random-surfer interpretation

Equation (2.3) admits a probabilistic reading: a surfer follows outgoing edges with probability d and teleports uniformly with probability $1 - d$ (Figure 2.2). The stationary distribution ranks nodes by long-run visit frequency. Damping prevents sinks from trapping all mass and guarantees primitivity of G_{PR} under mild conditions, so the Perron vector is unique ^[23].

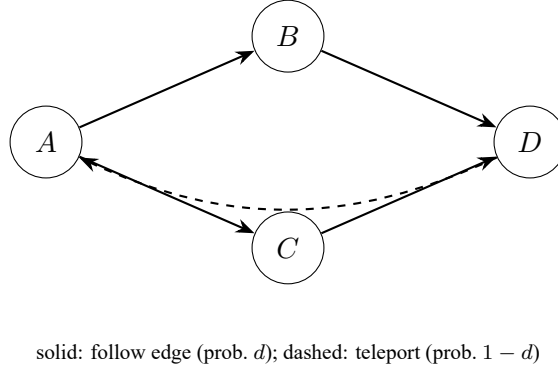


Figure 2.2: Random-surfer intuition for PageRank. Solid arrows are graph transitions; dashed arrows represent teleportation that mixes the chain.

2.5.2 Power iteration and complexity

In practice π is computed by power iteration:

$$\pi^{(t+1)} = G_{\text{PR}} \pi^{(t)}, \quad (2.4)$$

until $\|\pi^{(t+1)} - \pi^{(t)}\|_1 < \varepsilon$ or a maximum iteration count is reached. Each iteration costs $O(|E|)$ arithmetic operations for sparse graphs ^[49]. Therefore, methods that induce fewer edges—such as latest-allowance snapshots versus full transfer histories—reduce per-iteration cost and often memory. This complexity observation grounds hypothesis H3.

Default parameters in this research use damping $d = 0.85$ following common practice ^[15], with convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of 300 iterations as study-specific solver settings.¹

¹Tolerance and iteration caps are implementation choices for reproducible timing; they are not claimed as part of the

Robustness checks vary $d \in \{0.75, 0.85, 0.95\}$.

2.6 Reputation systems lineage

PageRank is one member of a family of spectral and propagation-based reputation mechanisms. Kleinberg’s HITS algorithm distinguishes hubs and authorities in hyperlinked corpora ^[50]. EigenTrust aggregates local trust scores in peer-to-peer networks with normalization to limit malicious inflation ^[16]. TrustRank propagates trust from a seed set of known-good nodes to combat web spam ^[17]. These systems share a theme: reputation is relational and should be hard to forge without costly endorsements from already-reputable nodes.

On-chain, Sybil attacks remain central ^[11]. Creating wallets is cheap; creating wallets that receive allowances or transfers from diverse counterparties is costlier. Sybil-adjusted stability proxies in this research (inbound counterparty ratio, tenure, active months) partially capture diversity of interaction but do not eliminate adversarial manipulation. Chapter 5 sketches a cost model for allowance-graph Sybils.

Machine-learning credit models on DeFi features ^[14] can outperform simple graph scores on labeled defaults when labels exist, but they trade interpretability for accuracy. EndorseRank and AWP prioritize transparent propagation rules suitable for audit ^[2].

2.7 ERC-20 allowances as endorsement

ERC-20 approve and EIP-2612 permit record explicit spending authorization ^{[18],[19]}. EndorseRank treats the latest allowance per owner–spender pair as edge strength on a directed endorsement graph and runs PageRank on that snapshot, eliminating historical time-decay because newer approvals override older ones.

Formally, for each token and owner–spender pair, let $a_{o \rightarrow s}$ be the decimal-adjusted latest allowance. Edges are aggregated across tokens by summing allowances into a single weight $w_{o \rightarrow s}$. The result-

classical PageRank formulation.

ing graph $G_{\text{approve}} = (V, E_{\text{approve}}, w)$ is typically much sparser than the transfer graph over the same wallets and window. PageRank on G_{approve} yields EndorseRank scores s_{ER} .

The endorsement interpretation is intentional but imperfect. Users approve routers for convenience, not always as a social judgment of the router’s “trustworthiness.” Contract addresses and EOAs coexist. Nevertheless, allowances are a clearer authorization signal than passive receipt of transfers, which may reflect market making, wash trading ^[29], airdrop farming ^[32], or routing without any endorsement intent.

2.8 What this research does not claim

This research does not claim that EndorseRank predicts trading profit, loss avoidance or liquidation. The research proposal planned to validate both methods against GMX V2 trading-success proxies and to bound that validation with an outcome-native reference (a PageRank on a star graph built from realized profit and loss). That reference is circular by construction: it is computed from the same PnL stream that defines the trading-success proxies, so its near-perfect agreement with the realized-gain proxy is the correlation of a variable with a monotone function of itself, not evidence about reputation. For the same reason, weak agreement between a social method and those proxies cannot be read as failure. The reference, the associated supplementary research question and claim, and the trading-success, inverse-risk and liquidation families are therefore withdrawn from the main narrative; their matrices are archived in Appendix 8.4 for completeness only. The remaining external check is a temporal holdout of future new owner–spender approvals (Chapters 3 and 4), which is out of window but of the same construct as the score.

2.9 Rank correlation theory

Reputation scores and validation proxies are continuous variables without a universal classification threshold. Following the AWP validation design^[3] and classical psychometrics^[22], this research evaluates monotonic and pairwise rank agreement, and it quantifies the sampling uncertainty of

every reported coefficient with bootstrap confidence intervals ^[34].

Spearman’s rank correlation ^[20] is Pearson correlation on ranks. For paired ranks (R_i, S_i) of n items,

$$\rho = 1 - \frac{6 \sum_{i=1}^n (R_i - S_i)^2}{n(n^2 - 1)} \quad (2.5)$$

in the absence of ties (with tie-aware variants used in software). Spearman’s ρ captures monotonic association.

Kendall’s τ ^[21] counts concordant and discordant pairs. For all pairs (i, j) with $i < j$,

$$\tau = \frac{2(C - D)}{n(n - 1)}, \quad (2.6)$$

where C (D) is the number of concordant (discordant) pairs, with tie corrections in the τ -b variant. Kendall’s τ is often more interpretable for pairwise ordering agreement and is the primary family-level summary in Chapter 4. Values are compared relatively across methods on the same cohort rather than against an uncited absolute threshold.

2.10 Theoretical framework

This study uses hyperlink–citation propagation, construct (domain) alignment, and computational complexity as the bases for comparing EndorseRank and AWP across reputation structure (H1), social-method alignment (H2), and computational efficiency (H3).

- Hyperlink–citation semantics: The original PageRank algorithm ^[15] treats inbound hyperlinks as citations. Do and Do^[4] apply this logic to Ethereum transfers; AWP^[3] uses time-decayed transfer edge weights. Figure 3.2 (Chapter 3) plots the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp(k(\Delta t - t_0))} \quad (2.7)$$

with $k = 0.01$ and $t_0 = 180$ days as used for AWP in this study. EndorseRank extends

citation-based propagation to ERC-20 approve/permit edges because allowances are explicit, on-chain delegations of spending authority ^{[18],[19]}. Unlike transfers, which may reflect trading or routing without trust intent, an allowance records explicit authorization that remains in force until revoked or replaced. Graph-based risk models ^[12] further show that network structure carries information beyond raw activity counts, which supports a directed endorsement graph in addition to transfer volume.

- Domain alignment framework: In measurement theory, construct alignment assesses how well an indicator tracks an intended construct ^[22]; Messick^[33] extends this to a unified judgment about what a score means and what its use implies. For rank-based reputation scoring, Spearman’s ρ and Kendall’s τ evaluate monotonic and pairwise ordering agreement without assuming a classification threshold, as in the AWP validation protocol^[3]. Contemporaneous transfer and allowance statistics check graph coherence; a temporal holdout of future new approvals is the out-of-window check.
- Computational complexity: PageRank’s per-iteration cost is $O(|E|)$ ^[15]. Allowance graphs are sparser than transfer graphs (fewer edges $|E|$), yielding lower per-iteration cost and reduced memory, and often fewer iterations in practice—the basis for H3. Because the solver is identical for both methods, any runtime difference is an expected consequence of edge sparsity, not an algorithmic contribution.

Figure 2.3 presents the conceptual model mapping the primary EndorseRank–AWP comparison to H1–H3 and RQ1–RQ4.

- Holdout, not outcome ranking: H1 and H3 concern EndorseRank and AWP only. H2/RQ2 compares contemporaneous allowance and transfer checks. RQ4 is a time-split test of future new owner–spender approvals on the spender cohort.

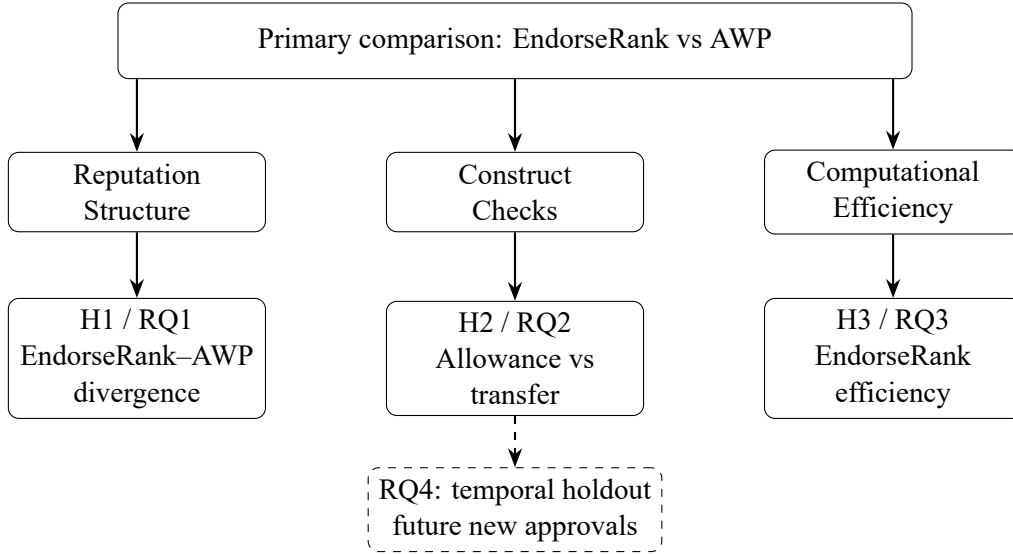


Figure 2.3: Conceptual model: primary EndorseRank vs. AWP comparison mapped to reputation structure (H1/RQ1), construct checks (H2/RQ2), computational efficiency (H3/RQ3), and the temporal holdout of future new approvals (RQ4).

2.11 Reader’s guide to terminology

The definitions in Chapter 1 establish formal constructs; the following guide clarifies how to interpret results and avoid common misreadings:

- Primary methods: EndorseRank (allowance graph) and AWP (transfer graph). Both are social PageRank methods on wallet-to-wallet edges; intended as deployable reputation signals.
- Primary graph construct: The graph a method is *designed* to rank (e.g., G_{approve} for EndorseRank). Distinct from checks applied after scoring.
- Contemporaneous check: Local transfer or allowance statistics computed in the same window as the score ^[22].
- Temporal holdout: Scores frozen at t_1 ; labels from t_2 new owner–spender approvals on inbound spenders.
- Matched wallet cohort: The primary same-window sample of $n = 5,521$ wallets with non-zero endorsement- and transfer-graph connectivity.

- Spender holdout cohort: $n = 1,335$ t1 inbound allowance recipients used for RQ4.
- Expanded wallet pool: A larger address set ($N = 99,080$) used for the node-count sensitivity of runtime beyond the matched cohort.

2.12 Domain alignment metrics

Following the AWP validation protocol^[3] and Cronbach and Meehl^[22], construct alignment for rank-based scores uses Spearman’s ρ and Kendall’s τ between reputation scores and check rankings. This research reports contemporaneous transfer and allowance checks and a temporal holdout of future new approvals; see Chapter 3 for operational definitions.

The literature reviewed here supports three design choices: (i) PageRank as an interpretable propagation backbone; (ii) allowances as endorsement edges distinct from transfers; and (iii) rank correlation plus a time split as the check design. These choices motivate the remaining sections of this chapter and, later, the operational pipeline in Chapter 3.

2.13 Worked example: PageRank on a four-node endorsement graph

To make Equation (2.3) concrete, consider four wallets $\{A, B, C, D\}$ with latest-allowance edges

$$A \rightarrow B (2), \quad A \rightarrow C (1), \quad B \rightarrow D (3), \quad C \rightarrow D (1),$$

and damping $d = 0.85$. Out-strengths are $d_A^{\text{out}} = 3$, $d_B^{\text{out}} = 3$, $d_C^{\text{out}} = 1$, $d_D^{\text{out}} = 0$ (dangling). After one power-iteration step from the uniform vector $\pi^{(0)} = \mathbf{1}/4$, mass flows preferentially toward D because both B and C endorse D , while A splits mass between B and C . Subsequent iterations mix in teleportation mass $(1 - d)/4 = 0.0375$ at every node. The stationary ranking places D highest, then B and C , then A —illustrating that *receiving* endorsements from nodes that themselves receive endorsements raises score, exactly the citation logic of PageRank^{[15],[23]}.

If the same wallets are connected by many small transfers instead of few allowances, the transition

matrix changes and the ranking can reorder even though the node set is identical. That is the microcosm of H1: EndorseRank and AWP need not agree.

2.14 DeFi risk taxonomy and where reputation fits

DeFi risk is multi-layered; Werner et al.^[8] organise it by protocol layer, and Zhou et al.^[27] systematise real-world incidents by attack surface. Smart-contract risk concerns bugs and upgrade keys^[51]. Oracle risk concerns manipulated or stale prices^[39]. Market risk concerns volatility and liquidation cascades^[52], including stablecoin and funding-market stress^[53]; the March 2020 crash illustrated how these layers interact^[9]. Governance risk concerns token-weighted capture^[54]. *Counterparty* or *behavioral* risk—the focus of this research—concerns whether a wallet’s historical interactions suggest reliable management of leveraged exposure.

On-chain reputation scores address only the behavioral layer, and only partially. They do not replace collateral, audits, or oracle design. Their role is to summarize relational structure (who authorizes whom; who pays whom) into an ordinal signal that can be audited and recomputed^[2]. EndorseRank targets the authorization facet of behavioral risk; AWP targets the payment-flow facet. Neither claims to predict exploits or governance attacks.

Artificial activity further complicates behavioral measurement. Wash trading inflates volume on decentralized exchanges^[28] and, at scale, on unregulated centralized venues^[29]; front-running and other forms of extractable value generate transfers that reflect ordering games rather than relationships^[55], and Qin et al.^[30] quantify how much of that value is realised on Ethereum. Scam tokens on Uniswap^[31] and cross-venue arbitrage flows^[56] likewise inflate transfer activity without durable counterparty relationships. Transfer graphs are especially exposed to artificial volume; allowance graphs are exposed to approval farming. Time-split checks and Sybil-oriented diagnostics are partial mitigations, not complete defenses.

2.15 Related measurement traditions

Beyond blockchain, reputation systems have been studied in e-commerce ^[57], peer-to-peer networks ^[16], and web spam ^[17]. Common lessons include: (i) reputation is relational; (ii) cheap identities enable Sybil attacks ^[11]; (iii) seed sets and normalization limit inflation; and (iv) evaluation requires external criteria, not only internal consistency ^[22]. EndorseRank inherits (i)–(iii) via PageRank damping and endorsement costs, and addresses (iv) via contemporaneous construct checks measured with Spearman’s ρ ^[20] and Kendall’s τ ^[21], and via a temporal holdout.

Personalized and topic-sensitive PageRank variants ^{[58],[59]} could, in future work, bias teleportation toward protocol-approved contracts or known-good wallets, analogous to TrustRank seeds. This research uses uniform teleportation for comparability with AWP^[3].

2.16 Accounts, gas, and why logs are the measurement surface

Ethereum-style accounts are either externally owned accounts (EOAs), controlled by private keys, or contract accounts, controlled by code ^[60]. Every state-changing interaction consumes gas and emits logs when contracts are written to do so. Reputation research that aims for auditability therefore privileges *logs* over off-chain indexes: logs are part of the canonical chain history and can be re-derived by any full node or public warehouse ^[61].

Gas costs shape the graphs we observe. On L2s, lower fees increase transfer frequency and approval churn, densifying G_{transfer} more than G_{approve} because payments are routine while approvals are intermittent setup actions. This economic asymmetry is one reason EndorseRank graphs are expected to remain sparse relative to transfer graphs (examined in Chapter 4).

2.17 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) are a well-known user-experience hazard: a compromised spender can drain tokens until the owner revokes. Security analyses of honeypot contracts

[62] and of phishing accounts on Ethereum [24] treat allowances as capabilities, not as neutral meta-data. EndorseRank interprets inbound allowances as endorsement of a spender’s capability, which is precisely why the construct is semantically distinct from inbound transfers. A wallet may receive many transfers without ever being trusted to spend on others’ behalf.

From a measurement standpoint, unlimited approvals create heavy-tailed edge weights. Decimal adjustment and optional winsorization can moderate extremes; this research sums decimal-adjusted latest allowances and relies on PageRank’s normalization by out-strength so that a single huge approval does not automatically dominate unless the owner has few outgoing endorsements.

2.18 AWP in depth: what the transfer-graph baseline measures

Do and Do^[4] apply PageRank and HodgeRank to Ethereum transfers as a social-credit signal. Adaptive Weighted PageRank (AWP), introduced by Do, Do, and Nguyen^[3], uses transfer edges whose weights combine value and recency through a logistic decay, and validates the ranking against inbound degree and inbound value—proxies for “how much economic attention a wallet receives.” Both the decay formulation (Equation (2.7)) and the in-degree/in-value proxies used in Chapters 3 and 4 come from the AWP paper. The framework is attractive because it is fully on-chain, rank-based, and free of off-chain identity.

EndorseRank preserves the PageRank-and-rank-correlation template but replaces the edge semantics. The empirical question is whether allowance edges yield a different, useful, and cheaper signal. Chapter 4 tests whether allowance edges produce a construct-distinct, allowance-aligned, and computationally cheaper score, and whether t_1 scores predict t_2 new approvals.

2.19 Comparison of graph reputation methods

Table 2.1 situates EndorseRank among related methods discussed in this chapter by edge semantics, temporal model, and validation design.

Table 2.1 shows that EndorseRank’s novelty is the choice of latest-allowance endorsement edges

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical validation
PageRank [15]	(web) Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust [16]	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank [17]	Links from seed goods	Static / seeded	Spam demotion
PageRank/HodgeRank on Ethereum [4]	Token transfers	Static window	Not reproduced in this thesis
AWP [3]	Token transfers	Logistic decay	In-degree / in-value
RiskProp [12]	Risk-bearing links	Propagation of risk	Account risk labels
EndorseRank (this study)	Latest allowances	Snapshot (override)	Allowance checks; t2 new approvals

as the primary social construct for DeFi wallets, not a new ranking algorithm.

2.20 Summary of the reviewed literature

Table 2.2 consolidates the sources that shape the design of this study. For each source it records the methodology, the strengths that this thesis borrows, the limitations that it tries to address, and the relation to the present work. The table is the evidence base for the research gap stated in Section 2.21.

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1998) [15]; Langville and Meyer (2006) [23]	Eigenvector ranking of a hyperlink graph; power iteration	Interpretable propagation; $O(E)$ per iteration; unique Perron vector under damping	Edge semantics fixed to hyperlinks; no notion of time	Solver and complexity argument reused unchanged for both compared methods

continued on next page

Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Kamvar et al. (2003) ^[16]	Trust aggregation in P2P networks; seeded propagation against spam	Normalisation and seed sets limit inflation; explicit adversary model	Requires explicit ratings or curated seeds; not on-chain	Motivates the Sybil-cost discussion and the uniform-teleportation baseline
Do and Do (2023) ^[4]	PageRank and HodgeRank on Ethereum transaction graphs as social credit	Framing of transfer connectivity as credit-relevant; fully public data	Transfers conflate payment and trust; no allowance semantics	Prior work on the transfer graph; distinguished from AWP in this thesis
Do, Do, and Nguyen (2023), AWP ^[3]	Value- and recency-weighted transfer PageRank with logarithmic decay; rank correlation against in-degree/in-value	Rank-based validation without default labels; recency handled explicitly	Proxies are computed from the same transfer data that builds the graph (mechanical alignment); dense graph	Baseline method, decay formula, and proxy protocol reproduced exactly on the same cohort
Lin et al. (2025), RiskProp ^[12]	De-anonymisation score plus network propagation of account risk	Shows structure carries information beyond activity counts; labelled evaluation	Needs risk labels; targets fraud, not trust or authorisation	Motivates refined edge semantics; RiskProp-style propagation kept as an archived extended baseline
Wu et al. (2022) ^[24]	Network embedding of Ethereum transaction graphs for phishing detection	Graph features outperform per-account features; large labelled set	Supervised; labels are phishing, not credit-worthiness	Supports the premise that graph position is informative; not a competitor method
Hassija et al. (2020) ^[13] ; Kotb (2023) ^[14]	Prospect-theory and machine-learning credit scoring on blockchain data	Directly target lending decisions	Require labels or off-chain features; limited interpretability	Contrast class: EndorseRank stays within auditable graph propagation

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Bellini et al. (2020) ^[25] ; Almasoud et al. (2020) ^[26]	Surveys of blockchain-based trust and reputation systems	Map the design space; identify explicit-rating and volume-based signals as dominant	Do not evaluate authorisation edges; no L2 measurement	Establish that latest-allowance edges are absent from the surveyed designs
Victor (2020) ^[40] ; Victor and Weintraud (2021) ^[28] ; Cong et al. (2023) ^[29]	Address clustering heuristics; wash-trading detection on DEXs and exchanges	Quantify how much on-chain activity is artificial or multi-address	Do not propose a reputation score	Explain why transfer-based signals are exposed to inflation; motivate the allowance construct
Cronbach and Meehl (1955) ^[22] ; Messick (1995) ^[33]	Construct validity in psychometrics	Separate what a score is designed to measure from external criteria	Not specific	Frame the contemporaneous checks as construct checks and the temporal holdout as external evidence
Efron (1979) ^[34] ; DiCiccio and Efron (1996) ^[35]	Bootstrap resampling and bootstrap confidence intervals	Distribution-free uncertainty for rank statistics; paired contrasts	Assumes exchangeable units	Provides the 95% intervals and $\Delta\tau$ contrasts reported in Chapter 4

2.21 Research gap

Three gaps follow from Table 2.2. First, every on-chain reputation method reviewed—PageRank/HodgeRank ^[4], AWP ^[3], RiskProp ^[12], and the systems catalogued in the surveys ^[25]—builds its graph from transfers or explicit ratings; none uses the ERC-20 allowance, the only on-chain event that records a deliberate delegation of spending authority. Whether a latest-allowance endorsement graph produces a different and useful ordering is therefore untested (addressed by O1 and RQ1). Second, the validation protocol of the transfer-graph baseline com-

pares scores with in-degree and in-value computed from the same transfer data that builds the graph, so agreement is partly mechanical; no study reports the same protocol on an allowance construct alongside its transfer counterpart with confidence intervals and paired difference tests on one cohort (O2 and RQ2). Third, reported runtimes for blockchain PageRank are rarely accompanied by edge counts, memory, and iteration counts under a fixed protocol, which makes efficiency claims hard to attribute to sparsity versus implementation (O3 and RQ3). A fourth, methodological gap concerns time: same-window graph statistics are routinely presented as validation even though they cannot show that a score anticipates future behaviour; a temporal holdout on future new approvals is the out-of-window check adopted here (O4 and RQ4).

2.22 Open problems motivating this research

Several open problems in on-chain reputation remain unresolved in the literature and motivate the present study:

1. Edge semantics: Which on-chain relations best encode trust or authorization under pseudonymity ^[2]?
2. Validation without defaults: How can methods be compared when loan-default labels are scarce ^[4]?
3. Computational cost: Can reputation scores refresh at interactive latencies as graphs grow ^[23]?
4. Time-split checks: How can evaluation avoid mistaking same-window graph statistics for out-of-window evidence ^[22]?
5. Adversarial robustness: What is the cost of forging reputation under cheap identities ^[11]?

EndorseRank addresses (1)–(4) with allowance edges, contemporaneous checks, runtime benchmarks, and a temporal holdout, and analyzes (5) at a model level in Chapter 5. Full empirical adversarial evaluation remains future work.

Chapter 3 turns the gap statement into an operational design: it fixes the data window and cohort, specifies how both graphs are constructed and solved with one shared PageRank implementation,

defines the proxy families and the temporal holdout, and sets the timing and bootstrap protocols whose outputs Chapter 4 reports.

3 Research Methodology

Chapter 2 closed with the research gap: no reviewed method ranks wallets on allowance edges, the transfer-graph baseline validates against proxies drawn from its own input data, and efficiency claims are rarely tied to edge counts under a fixed protocol. This chapter specifies the empirical design used to compare EndorseRank and Adaptive Weighted PageRank (AWP) as social reputation scores on Arbitrum One so that each of those gaps is addressed by a measurable procedure. The design covers notation, data sources, a two-phase collection strategy, pipeline architecture, event decoding, graph construction algorithms, contemporaneous construct checks, a temporal holdout, statistical tests with bootstrap inference, computational benchmarks, scaling and sensitivity extensions, reproducibility, and ethics. The primary comparison is between EndorseRank on the ERC-20 allowance graph and AWP on the time-decayed transfer graph ^[3], both solved with classical PageRank ^[15].

The methodological stance follows construct (domain) alignment: reputation scores are continuous rankings, and checks use observable wallet-level statistics rather than binary default labels ^[22]. In measurement terms, each ranking method is an indicator of an unobserved reputation construct. Contemporaneous transfer and allowance statistics check graph coherence. A temporal holdout of future new owner–spender approvals is the out-of-window check. Alignment statistics ask whether the ordering induced by a method co-varies with the ordering induced by each check on the same wallets. This approach is appropriate when the scientific claim is comparative—EndorseRank versus AWP—rather than prediction of trading profit or credit default.

All procedures operate on publicly available blockchain logs within a fixed observation window, with deterministic seeds and an open evaluation pipeline so that reported tables can be regenerated from cached parquet. The chapter is organized to move from symbols and data acquisition, through graph and proxy formalisms, to evaluation statistics, computational protocol, and research ethics.

Readers primarily interested in empirical outcomes may skim Sections 3.1–3.5 and focus on Sections 3.6–3.10; readers implementing a replication should follow the full pipeline description and Section 3.13.

Four principles govern every operational choice.

- Same-seed comparability: EndorseRank and AWP are always scored on identical wallet sets so that differences cannot be attributed to sampling inconsistency.
- Pre-registration by configuration: observation window, topic hashes, damping, decay parameters, bootstrap resamples and seed, and benchmark seeds are fixed in configuration before interpreting results.
- Separation of construction and timing: graph construction is performed once per cohort; computational benchmarks time only the PageRank solve.
- Primary scope: claims C1–C4 and hypotheses H1–H3 concern EndorseRank and AWP. RQ4 uses the spender holdout cohort.

3.1 Notation

Table 3.1 summarizes the operational symbols used throughout this chapter and in Chapters 4–5 (a compact front-matter summary appears in Table 2). Vectors are bold; sets are calligraphic when helpful. Wallet addresses are treated as opaque identifiers and are never linked to off-chain identities.

Unless otherwise stated, PageRank is the weighted formulation of Page et al.^[15]: at iteration t ,

$$\mathbf{r}^{(t+1)} = (1 - d) \boldsymbol{\pi} + d P^\top \mathbf{r}^{(t)} + d (\mathbf{r}^{(t)\top} \mathbf{1}_{\text{dang}}) \boldsymbol{\pi}, \quad (3.1)$$

where P is the row-stochastic transition matrix induced by positive edge weights, $\boldsymbol{\pi}$ is the uniform teleportation distribution over nodes, and $\mathbf{1}_{\text{dang}}$ marks dangling nodes (zero out-strength). Explicitly, if $w(u, v)$ denotes the weight of the directed edge $u \rightarrow v$ and $s(u) = \sum_v w(u, v)$ is the out-strength,

Table 3.1: Notation for graphs, scores, proxies, and parameters.

Symbol	Meaning
$\mathcal{W}$	Matched wallet cohort ($ \mathcal{W} = n = 5,521$)
$\mathcal{W}_{\text{pool}}$	Expanded wallet pool ($ \mathcal{W}_{\text{pool}} = N = 99,080$)
$G_{\text{approve}} = (V_A, E_A, w_A)$	EndorseRank allowance graph
$G_{\text{transfer}} = (V_T, E_T, w_T)$	AWP transfer graph
$w_A(u, v)$	Latest allowance weight, owner $u \rightarrow$ spender v
$w_T(u, v)$	Time-decayed transfer weight, sender $u \rightarrow$ receiver v
$\mathbf{s}_M$	Reputation score vector for method M
$\mathbf{p}_P$	Proxy value vector for proxy P
d	PageRank damping factor (default 0.85)
τ_{tol}	PageRank convergence tolerance (10^{-8})
I_{max}	Maximum PageRank iterations (300)
$\sigma(\Delta t)$	Logistic time-decay on transfer age
k, t_0	Decay steepness (0.01) and midpoint (180 days)
Δt	Age of an event in days before observation end
ρ	Spearman’s rank correlation
τ	Kendall’s rank correlation
T_{obs}	Observation end timestamp (2026-05-31 UTC)

then

$$P_{u,v} = \begin{cases} w(u, v)/s(u) & \text{if } s(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Dangling mass is redistributed uniformly through the teleportation term, which is the standard treatment in sparse web and transaction graphs. Iteration stops when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \tau_{\text{tol}}$ or after I_{max} iterations. Scores are renormalized to sum to one over reachable nodes. Dense ranks for alignment are assigned by sorting scores descending (rank 1 = highest score), with wallet address as a deterministic tie-break when scores are equal.

The damping factor d interpolates between pure graph propagation ($d \rightarrow 1$) and uniform scores ($d \rightarrow 0$). The default $d = 0.85$ follows the classical web setting and the AWP baseline^[3]. Robustness sweeps vary d while holding edge weights fixed (Section 3.12). Tolerance $\tau_{\text{tol}} = 10^{-8}$ and $I_{\text{max}} = 300$ are intentionally strict relative to interactive analytics settings, so that runtime differences reflect graph structure rather than early stopping.

3.2 Data sources and sampling

This section fixes the empirical universe: which chain, which time window, which public tables, which event types, and which wallets enter the primary sample. Downstream graph construction and proxy formulas never invent events outside this universe.

- Primary chain: All on-chain events are drawn from Arbitrum One (chain ID 42161), an Ethereum Layer-2 rollup with dense DeFi activity and public log availability ^[1]. Restricting to a single chain avoids cross-chain address aliasing—the same private key controlling addresses on multiple chains would otherwise appear as unrelated nodes—and keeps gas-cost and latency regimes comparable across wallets. The chain is the sole empirical venue for this comparison.
- Observation window: The study window is 2025-12-01 00:00:00 UTC through 2026-05-31 23:59:59 UTC (six calendar months). Exclusive end timestamps in SQL use 2026-06-01 00:00:00 UTC so that the inclusive calendar end is 2026-05-31. This window is long enough for monthly-batched reputation extracts under BigQuery byte budgets and short enough that latest-allowance snapshots remain contemporaneous with transfer histories. AWP time-decay is anchored at observation end $T_{\text{obs}} = 2026-05-31\ 23:59:59\ \text{UTC}$, so $\Delta t = 0$ for events at the window’s close and $\Delta t \approx 181$ days for events at the window’s open.
- BigQuery source: Event logs are read from the public dataset

`bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

Each row corresponds to a contract log entry with block timestamp, emitting address, topics, and data. Queries filter on `block_timestamp`, contract address (when applicable), and event topic hashes. The public dataset removes the need for a private archive node while remaining auditable: any researcher with Google Cloud access can re-issue the same SQL against the same table. SQL templates, dry-run cost guardrails, and acquisition metadata are documented in Appendix 8.1.

- Reputation events: EndorseRank uses ERC-20 Approval logs ^[18], including allowances established via EIP-2612 permit flows that ultimately emit the same Approval event ^[19]. AWP uses ERC-20 Transfer logs. Both streams are wallet-filtered to the cohort under study: a log is retained if the owner or spender (approvals) or the sender or recipient (transfers) lies in the seed wallet set. Filtering at query time reduces scanned payload relative to full-chain extracts and focuses the reputation subgraph on economically relevant neighbors of the cohort. Extracts run in six monthly batches to respect per-query byte caps and to allow resumable checkpoints if a month fails.
- Matched cohort definition: The primary alignment sample is wallets with non-zero connectivity in the reputation subgraph induced by approval or transfer edges in the observation window. After BigQuery extraction and preprocessing, this matched wallet cohort has size $n = 5,521$. Both EndorseRank and AWP are scored on the same seed set so that differences in alignment and runtime are attributable to ranking logic and graph structure, not to inconsistent sampling.
- What is not in scope: The study does not ingest mempool data, off-chain order books, centralized-exchange account maps, or cross-chain bridges. It does not attempt entity resolution across multiple addresses controlled by one actor. The public log table is authoritative, and decoding is performed in the open pipeline.

3.3 Two-phase collection strategy

Data acquisition and evaluation proceed in two phases that separate expensive remote scans from local, repeatable analysis. The separation is intentional: once parquet artifacts exist, alignment tables, benchmarks, and robustness sweeps can be regenerated without further cloud cost, which is essential for third-party replication and reanalysis.

- Phase 1 (reputation streams): Extract wallet-filtered ERC-20 Approval and Transfer events in monthly batches. The matched wallet cohort is the subset with non-zero approval or trans-

fer connectivity ($n = 5,521$). Each month is an independent query with its own dry-run estimate and checkpoint, so a failure in one month does not invalidate prior months. Preprocess approvals to latest allowances per owner–spender–token: given multiple approvals for the same triple, only the last timestamp in the window survives, and a final zero allowance removes the edge. Preprocess transfers to decimal-adjusted event tables with timestamps, preserving sender, recipient, value, token, and block time. These tables are the sole inputs to EndorseRank and AWP graph construction for the primary comparison.

- Phase 2 (local evaluation): From cached parquet, compute reputation ranks, transfer and allowance checks, Spearman and Kendall alignment, computational benchmarks, and scaling/robustness extensions. Supplemental active wallets for the expanded pool are drawn from reputation-subgraph addresses (and optional supplemental parquet) without requiring a full-chain re-scan for every robustness experiment. No additional BigQuery scans are required once the Phase 1 extracts and any supplemental active-wallet parquet are available.
- Cost guardrails: Every remote query begins with a mandatory dry-run estimate of bytes processed before extraction. Each query is aborted if the estimate exceeds a 150 GiB per-query cap.¹ Operators must pass an explicit confirmation flag before paid scans proceed. Extraction metadata—wallet counts, row counts, bytes processed, and timestamps—is logged in the study’s acquisition record (Appendix 8.1). These controls implement responsible use of public cloud resources and make the acquisition cost of the study auditable.

3.4 Pipeline architecture

Figure 3.1 summarizes the end-to-end flow from public logs to the research tables. Remote stages write immutable parquet; local stages are idempotent given those inputs and a single versioned configuration file.

Operationally, Phase 1 scripts write monthly approval/transfer shards and consolidated latest-

¹The per-query byte cap is an operational policy of this study’s acquisition pipeline, not a property of BigQuery itself.

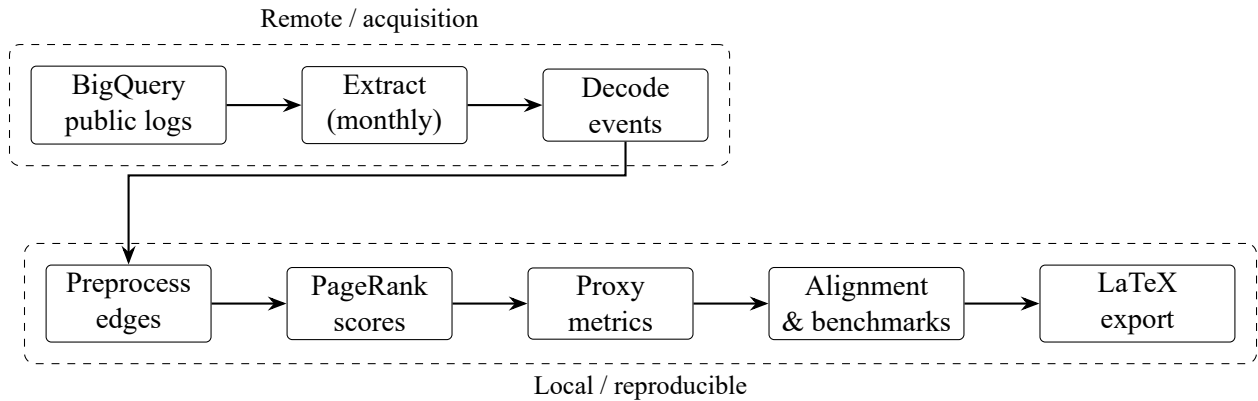


Figure 3.1: Evaluation pipeline: public Arbitrum logs are extracted and decoded, reputation edges are preprocessed, EndorseRank and AWP ranks are computed, checks and alignment statistics are evaluated, and LaTeX result fragments are exported.

allowance and transfer-event tables; Phase 2 scripts read those tables, build graphs, run PageRank, compute checks, and export the result tables reported in Chapter 4. The same pipeline supports offline verification on synthetic fixtures without cloud credentials.

3.5 Event decoding

Blockchain logs store events as topics and a data payload rather than as named fields. Decoding maps each log row to a typed record used by graph construction and construct checks. Incorrect decoding would silently corrupt both reputation graphs; this section therefore records the exact topic hashes and field mappings used in the pipeline.

- **Log anatomy:** An Ethereum-compatible log contains an emitting contract address, an ordered list of topics, and an ABI-encoded data blob. Topic 0 is the keccak-256 hash of the event signature string, which uniquely identifies the event type among well-known standards. Indexed parameters occupy subsequent topics as 32-byte words (addresses are left-padded); non-indexed parameters occupy the data field in ABI order^[18]. BigQuery exposes these fields as columns; the pipeline applies topic filters in SQL and performs ABI decoding in Python for fields that require numeric interpretation.

- ERC-20 Approval: Topic 0 is

`0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925,`

the hash of `Approval(address,address,uint256)`. Topics 1 and 2 hold the owner and spender addresses (left-padded); the allowance amount is ABI-decoded from the data field and scaled by the token’s decimals. Unlimited approvals (`type(uint256).max`) are retained as large finite weights after decimal adjustment; they contribute strong endorsement edges, which is intentional: an unlimited approval is a strong on-chain authorization. EIP-2612 `permit` does not introduce a separate reputation event: successful permits emit the same `Approval` log, so `EndorseRank` observes the resulting allowance snapshot regardless of whether authorization was set by an on-chain approve call or an off-chain signature ^[19].

- ERC-20 Transfer: Topic 0 is

`0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef,`

the hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed; value is decimal-adjusted. Mint and burn patterns that use the zero address are retained if they touch cohort wallets, but they are rare relative to wallet-to-wallet transfers in the filtered extract. Self-transfers are retained in the event table but excluded from Sybil-stability inbound-counterparty calculations (Section 3.9), because a wallet transferring to itself does not evidence an external counterparty relationship.

- Address normalization and units: All addresses are lower-cased before joins so that checksummed and non-checksummed hex forms collide correctly. Token amounts are stored as floating-point decimal-adjusted values for weighting; ranking uses relative magnitudes within a cohort rather than absolute USD conversion for allowance and transfer edges.

3.6 Graph construction

`EndorseRank` and `AWP` share the PageRank solver in Equation (3.1) but differ in edge semantics. Both methods induce a directed, weighted graph on wallet nodes, restrict attention to the subgraph

touching the seed wallet set (edges with at least one endpoint in the cohort, including one-hop neighbors), and emit a score for every cohort wallet (missing nodes receive score 0 before ranking). Subgraph restriction is important for scalability and for interpretability: the score of a cohort wallet may receive mass from non-cohort neighbors that endorse or transfer to it, but the evaluation reports scores only on $\mathcal{W}$.

The conceptual distinction is endorsement versus flow. An allowance is an explicit, revocable authorization to spend; a transfer is a completed movement of value that may reflect payment, routing, market-making, or wash activity without a trust commitment ^{[4],[18]}. EndorseRank therefore treats the latest allowance snapshot as a citation-like endorsement graph in the sense of Page et al.^[15]. AWP treats historical transfers as time-decayed citations, down-weighting older flows so that recent economic activity dominates.

3.6.1 EndorseRank allowance graph

Let $\mathcal{A}$ be the set of latest allowance records after preprocessing: for each token and owner–spender pair, only the chronologically last Approval in the observation window is retained (zero allowances remove the edge). Across tokens, weights are summed:

$$w_A(u, v) = \sum_{\text{token}} a^*(u, v, \text{token}), \quad (3.3)$$

where a^* is the latest decimal-adjusted allowance. The directed edge $u \rightarrow v$ encodes owner u endorsing spender v . PageRank damping is $d = 0.85$. No temporal decay is applied: allowance override semantics replace time-based weighting, because a newer approval supersedes an older one for the same pair ^[18].

Algorithm 1 formalizes construction and scoring.

Algorithm 1 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$, seed wallets $\mathcal{W}$, damping d , tolerance τ_{tol} , max iterations I_{max}

Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{W}$

```
1:  $E \leftarrow \emptyset$ 
2: for each  $(u, v)$  in unique owner–spender pairs in  $\mathcal{A}$  do
3:    $w \leftarrow \sum_{\text{token}} a^*(u, v, \text{token})$ 
4:   if  $w > 0$  then
5:     add edge  $(u \rightarrow v, w)$  to  $E$ 
6:   end if
7: end for
8:  $E \leftarrow$  edges in  $E$  incident to at least one wallet in  $\mathcal{W}$ 
9:  $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, d, \tau_{\text{tol}}, I_{\text{max}})$ 
10: return  $\mathbf{s}_{\text{ER}}[w] \leftarrow \mathbf{s}[w]$  for  $w \in \mathcal{W}$  (else 0)
```

3.6.2 AWP transfer graph

AWP follows the IEEE RIVF 2023 transfer-graph method^[3]: each transfer of value v at age Δt days before observation end T_{obs} contributes a decayed weight

$$\sigma(\Delta t) = \frac{1}{1 + \exp(k(\Delta t - t_0))}, \quad k = 0.01, \quad t_0 = 180 \text{ days}. \quad (3.4)$$

Edge weights aggregate over transfers:

$$w_T(u, v) = \sum_{e: u \rightarrow v} v_e \sigma(\Delta t_e). \quad (3.5)$$

Damping is again $d = 0.85$. Figure 3.2 plots $\sigma(\Delta t)$ over the relevant age range: recent transfers retain nearly full weight, while transfers older than the midpoint t_0 are smoothly down-weighted.

Algorithm 2 formalizes construction and scoring.

Because G_{approve} records durable authorizations and G_{transfer} records historical value flows, the two graphs differ in density and temporal sensitivity. Approvals are relatively rare, high-intent events; transfers are frequent and often mechanical. On the matched cohort, the approval subgraph is substantially sparser than the transfer subgraph (Chapter 4), which is the structural basis for the

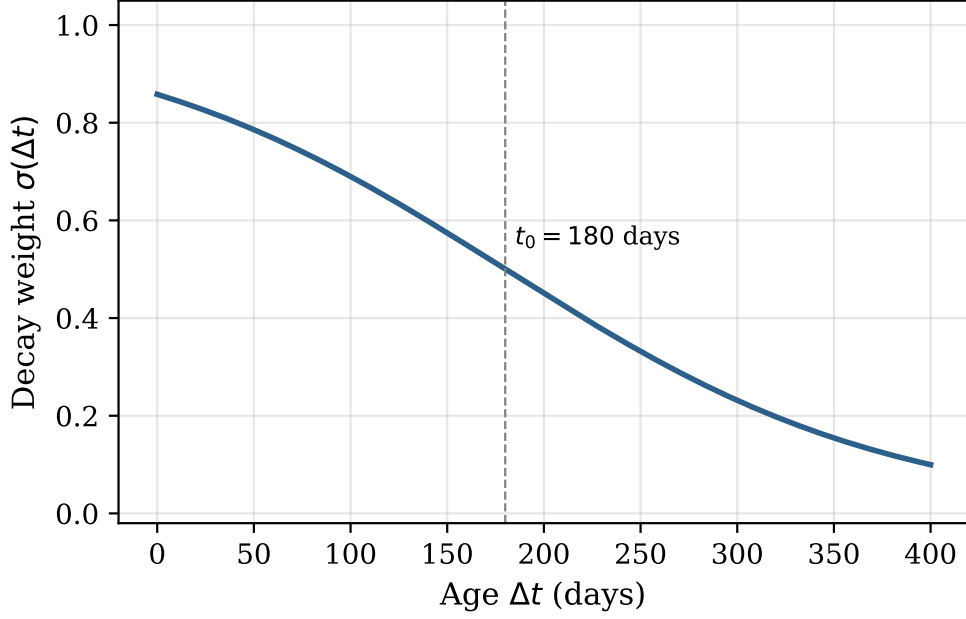


Figure 3.2: Logistic time-decay $\sigma(\Delta t)$ used by AWP ($k = 0.01$, $t_0 = 180$ days). Horizontal axis is event age in days before 2026-05-31 UTC; vertical axis is the multiplicative weight applied to transfer value.

Algorithm 2 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$, seed wallets $\mathcal{W}$, observation end T_{obs} , parameters $k, t_0, d, \tau_{\text{tol}}, I_{\text{max}}$

Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{W}$

- 1: $E \leftarrow \emptyset$
 - 2: **for** each transfer $e = (u, v, v_e, t_e)$ in $\mathcal{T}$ **do**
 - 3: $\Delta t \leftarrow (T_{\text{obs}} - t_e)$ in days
 - 4: $w_e \leftarrow v_e \cdot \sigma(\Delta t)$ with σ as in Equation (3.4)
 - 5: accumulate w_e on edge $(u \rightarrow v)$
 - 6: **end for**
 - 7: remove non-positive edges; retain edges incident to $\mathcal{W}$
 - 8: $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, d, \tau_{\text{tol}}, I_{\text{max}})$
 - 9: **return** $\mathbf{s}_{\text{AWP}}[w] \leftarrow \mathbf{s}[w]$ for $w \in \mathcal{W}$ (else 0)
-

computational-efficiency hypothesis (H3): per-iteration PageRank cost scales with $|E|$, so fewer edges imply faster solves and lower memory for the adjacency structures in Equation (3.2).

- Implementation notes shared by both methods: Edge lists are stored as triples (`from_node`, `to_node`, `weight`) in memory. Duplicate edges are aggregated by summing weights before the solve. Non-positive weights are dropped. The solver builds a sparse compressed-row transition matrix, iterates Equation (3.1), and returns a dictionary from node identifier to score. Cohort scores are aligned to $\mathcal{W}$ with missing entries set to zero, then converted to dense ranks for correlation.

3.7 Temporal holdout protocol

Scores for the holdout are computed from events with timestamp at or before $T_1 = 28$ February 2026, 23:59:59 UTC. Labels use events from $T_2^{\text{start}} = 1$ March 2026 through $T_2^{\text{end}} = 31$ May 2026. The claimed label is the count of new owner–spender approval pairs in T_2 that are absent from the latest-positive t1 snapshot, attributed to the spender. The claimed cohort is t1 inbound spenders ($n = 1,335$).

Other t2 quantities (revoke rate, an approval-then-outbound drain heuristic, Aave borrower liquidations) were computed in the pipeline. They are not reported as supporting evidence: revoke and drain had the wrong sign relative to an endorsement reading, and Aave events on this spender set were empty or the wrong construct.

3.8 Construct-check design

Primary evaluation compares EndorseRank and AWP only. For each method M and each check P , the pipeline computes scores $\mathbf{s}_M$ and check values $\mathbf{p}_P$, then reports Spearman’s ρ and Kendall’s τ with 95% bootstrap intervals (Section 3.10). Checks are oriented so that higher values imply higher expected reputation before correlation, following the AWP validation protocol^[3].

Two contemporaneous families structure the same-window comparison:

1. Transfer (AWP baseline proxies^[3]): inbound transfer activity.
2. Allowance: inbound approval activity (wallet as spender).

They are *graph-adjacent*: they are computed from the same event classes that build G_{transfer} and G_{approve} , but they are local statistics rather than global PageRank scores. Strong alignment of AWP with transfer checks, or of EndorseRank with allowance checks, is within-construct coherence, not independent predictive power. Sybil-adjusted activity summaries may appear in robustness tables; they are not headline claims.

The out-of-window check is the temporal holdout in Section 3.7. Packin and Lev-Aretz^[2] caution that decentralized credit scores can become opaque; publishing explicit formulas (Section 3.9) and separating same-window checks from the time split counters that opacity.

3.9 Operational definitions of proxies

Let $\mathcal{W}$ be the matched cohort. Notation below matches the check-computation module of the open evaluation pipeline.

3.9.1 Transfer family

Let $\mathcal{T}_{\text{in}}(w)$ be inbound transfers to wallet w (any sender). Define

$$\text{in_degree}(w) = |\{ u : \exists \text{ transfer } u \rightarrow w \}|, \quad (3.6)$$

$$\text{in_value}(w) = \sum_{e: \cdot \rightarrow w} v_e. \quad (3.7)$$

These are the domain-intuition baselines of the AWP paper^[3]: economically important addresses tend to receive transfers from more counterparties and in larger aggregate value. Unlike PageRank, in_degree and in_value are strictly local: they ignore the reputation of senders. Comparing AWP (global, time-decayed) to these local baselines tests whether propagation and recency weighting add ordering information beyond raw inflows.

3.9.2 Allowance family

Let $\mathcal{A}_{\text{in}}(w)$ be latest allowances in which w is the spender (owner $\rightarrow w$). Define

$$\text{in_approve_degree}(w) = |\{u : a^*(u, w, \cdot) > 0\}|, \quad (3.8)$$

$$\text{in_approve_value}(w) = \sum_{u, \text{token}} a^*(u, w, \text{token}). \quad (3.9)$$

These checks measure endorsement received, the construct EndorseRank is designed to rank. Strong alignment of EndorseRank with this family is expected and is interpreted as within-construct validity. As with transfer baselines, the allowance checks are local: they count and sum inbound approvals without propagating through multi-hop endorsement chains. Out-of-window evidence is reserved for the temporal holdout.

3.9.3 Sybil-adjusted stability family

Self-transfers ($u = w$) are excluded from inbound counters. Let $N_{\text{tx}}(w)$ be the number of inbound transfers to w from other addresses and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Let $t_{\text{min}}(w)$ and $t_{\text{max}}(w)$ be the first and last timestamps at which w appears as sender or recipient in the transfer table, and let $M(w)$ be the number of distinct calendar months of activity (inbound months and full-span months, taking the maximum). Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \quad (3.10)$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\text{max}}(w) - t_{\text{min}}(w)}{86400} \quad (\text{seconds to days}), \quad (3.11)$$

$$\text{active_months}(w) = M(w). \quad (3.12)$$

High counterparty ratio penalizes wallets that receive many transfers from few addresses (a simple Sybil/wash pattern). Tenure and active months capture persistence of on-chain activity within the observation window. These measures are imperfect Sybil defenses—coordinated adversaries can

still diversify counterparties—but they operationalize the intuition that durable, multi-counterparty activity is more reputation-relevant than bursty self-dealing ^[2].

3.10 Statistical tests

Reputation scores and validation proxies are continuous rank variables, not binary default labels. Following the AWP validation protocol^[3] and the construct-validity tradition of Cronbach and Meehl^[22], alignment is measured with rank correlations that are invariant to monotonic transforms of either variable, and every reported coefficient carries a bootstrap confidence interval.

- Spearman’s ρ : Let R_i and S_i be the dense ranks of score and proxy for wallet $i = 1, \dots, n$. Spearman’s coefficient is the Pearson correlation of these ranks:

$$\rho = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 \sum_{i=1}^n (S_i - \bar{S})^2}}. \quad (3.13)$$

Spearman captures monotonic agreement: if higher reputation consistently co-occurs with higher proxy values, ρ is positive even when the relationship is nonlinear in the raw scores.

- Kendall’s τ : Kendall’s coefficient counts concordant and discordant pairs. For all pairs $i < j$,

$$\tau = \frac{2(C - D)}{n(n - 1)}, \quad (3.14)$$

where C is the number of pairs ordered the same way by both rankings and D is the number ordered differently (ties handled by the implementation’s standard tie correction). Kendall’s τ is more interpretable as a pairwise ordering probability and is less sensitive to large rank displacements than Spearman in some heavy-tailed settings.

- Why both: Spearman summarizes global monotonic association; Kendall emphasizes pairwise order fidelity. Reporting both follows the AWP protocol^[3] and allows readers to verify that family-level conclusions are not an artifact of a single rank statistic. In blockchain reputation settings, score distributions are often heavy-tailed: a few hubs receive large mass while

many wallets share near-zero scores. Rank correlations remain well-defined in that regime, whereas Pearson correlation on raw scores would be dominated by hubs.

- Family aggregation: Family-level summaries use mean τ across proxies within a family. For example, the transfer-family mean is $\frac{1}{2}(\tau_{\text{in_degree}} + \tau_{\text{in_value}})$. Method comparisons on the same cohort and proxies are relative, not absolute claims about universal reputation. Absolute τ cutoffs are not used as decision rules; confidence intervals are.
- Uncertainty: sampling uncertainty is quantified with a paired wallet bootstrap^[34]. The matched cohort is resampled with replacement $B = 400$ times (seed 42, fixed in configuration); the same index matrix is used for every method and proxy so that resampled statistics are paired across methods. For each proxy τ , each family mean, and the inter-method τ , the 2.5th and 97.5th percentiles of the bootstrap distribution form a 95% percentile interval^[35]. Because the population Kendall τ is a U-statistic, the percentile interval is first-order accurate at $n = 5,521$; the small number of resamples was chosen to keep the $O(B n \log n)$ cost of the tie-corrected τ tractable across all methods and proxies, and it limits the precision of the interval end-points to roughly ± 0.005 .
- Difference tests: the comparisons on which the argument rests are pre-specified as paired contrasts and tested with the same resamples: the bootstrap distribution of $\Delta\tau = \tau_A - \tau_B$ is computed resample by resample, and its 95% percentile interval is reported. A contrast is called a difference when the interval excludes zero. The pre-specified contrasts are (i) EndorseRank on its allowance family minus EndorseRank on the transfer family; (ii) AWP on the transfer family minus EndorseRank on the transfer family; (iii) EndorseRank on the allowance family minus AWP on the allowance family; (iv) AWP minus EndorseRank on the Sybil-stability family; (v) AWP minus EndorseRank on the GMX trading-success proxies (reported for completeness only); and (vi) EndorseRank on `in_approve_degree` minus EndorseRank on `in_degree`. Paired resampling removes the between-resample variance shared by the two coefficients, so the contrast interval is narrower than the difference of the two marginal intervals would suggest.

- Orientation and missingness: Proxies are pre-oriented so that larger values indicate better reputation. Wallets absent from a proxy source receive zeros after alignment to $\mathcal{W}$, which is conservative for sparse activity and is applied identically to both methods. No wallet is dropped from one method but retained in the other: the correlation sample is always the full matched cohort for primary tables.
- What is not tested: The study does not claim causal identification (reputation does not “cause” future approvals in an interventional sense), does not fit predictive classifiers with train/test splits, and does not use p -values as the decision criterion; interval exclusion of zero for a pre-specified contrast is the decision rule. The scientific target is comparative construct alignment and computational cost under a fixed, transparent protocol.

3.11 Computational benchmark (Claim C1)

Claim C1 concerns computational efficiency of EndorseRank versus AWP on identical inputs. The benchmark protocol isolates PageRank solve cost from one-time graph construction, because construction cost is dominated by I/O and preprocessing that would be amortized in a deployed scorer that updates incrementally, whereas the iterative solve is the recurring computational core of both methods.

- Timed quantity: For each method, the weighted edge list is built once per cohort outside the timed loop; extraction, decoding, allowance collapsing and transfer decay are therefore excluded. The timed section is the solver call, which assembles the sparse transition operator from that edge list and runs power iteration with damping $d = 0.85$, tolerance $\tau_{\text{tol}} = 10^{-8}$, and at most $I_{\text{max}} = 300$ iterations. Both parts of the timed section are linear in $|E|$, and both would be repeated at every refresh of a deployed scorer. These numerical settings match the configuration used for alignment scores, so runtime and alignment refer to the same mathematical object.
- Timing protocol: Every configuration is measured in one session with a single protocol. One

untimed warm-up solve is run first; it populates caches, triggers just-in-time paths, and is the only run instrumented for peak memory (memory tracing is not active during timed runs because it would inflate wall-clock time). Five timed solves follow. Reported statistics are the mean and standard deviation of the five wall-clock times, the mean iteration count, the warm-up peak memory, and the node and edge counts of the pre-built graph. Measurements are memoized within the session by a key consisting of the edge multiset signature, damping, tolerance, iteration cap, and repeat count; wherever the same graph is tabulated under the same settings—the primary benchmark, the $d = 0.85$ row of the damping sweep, the in-cohort scaling stage with $n = 5,521$, and the full expanded-pool stage, whose induced subgraph coincides with the cohort graph (Section 3.12.1)—the tables therefore show identical numbers by construction rather than independent measurements of the same object. Edge counts are reported in every runtime table because they are the leading structural explanation for runtime differences under $O(|E|)$ per-iteration complexity ^[15]. Per-run timings are retained in the machine-readable summary (Section 3.13).

- Cohorts: The primary benchmark uses the matched wallet cohort ($n = 5,521$). Node-count sensitivity uses deterministic subsamples of the expanded wallet pool (Section 3.12).
- Fairness of comparison: Both methods use the same solver, tolerance, iteration cap, damping (unless a sensitivity sweep varies d), and seed wallet set. Differences therefore reflect graph density and weight structure—allowance snapshots versus time-decayed transfers—rather than implementation asymmetry. Hardware and software versions are recorded in the evaluation manifest so that absolute seconds can be interpreted relative to the experimental machine; speedup ratios (AWP time divided by EndorseRank time) are the preferred comparative summary, and they are always reported next to the corresponding edge ratio.

3.12 Scaling and robustness extensions

Beyond the matched cohort, the study evaluates runtime sensitivity to node count on a larger address set and stress-tests alignment under parameter and sample perturbations. These extensions support H3 (efficiency at larger n) and characterise how far family-level conclusions depend on the sample definition. They are not alternative primary samples: unless otherwise noted, alignment claims in Chapter 4 refer to the matched cohort, and node-count sensitivity refers to the expanded pool.

3.12.1 Expanded wallet pool and node-count sensitivity of runtime

The expanded wallet pool $\mathcal{W}_{\text{pool}}$ unions the matched cohort (5,521 addresses) with 93,559 supplemental addresses sampled from monthly Arbitrum One ERC-20 activity in the observation window (the supplemental active-wallet parquet produced by the sampling script), yielding $N = |\mathcal{W}_{\text{pool}}| = 99,080$ addresses in this study. Runtime benchmarks at intermediate sizes use deterministic SHA256 subsampling with seed string `benchmark-tier2-v1`: wallets are ordered by

SHA256(seed || address) with seed `benchmark-tier2-v1`

and the first n addresses form the subsample. The seed string is a fixed configuration constant. Reported stages are $n = 10,000$, $n = 50,000$ and the full pool $n = N = 99,080$, always on the same approval and transfer parquet. In-cohort stages with $n \leq 5,521$ are reported in Appendix 8.1. Deterministic hashing avoids the non-reproducibility of pseudorandom draws across library versions.

Two properties of this design must be stated because they bound what the experiment can show. First, at every stage the graph is the subgraph of the cohort-window approval and transfer parquet induced by the sampled wallets, so $|E|$ at a stage is determined by which wallets the hash order selects and is not monotone in n by construction. Second, the supplemental pool addresses were meant to be accompanied by a second-tier extraction of their own approval and transfer logs; that extraction did not complete (Appendix 8.1), so the full-pool stage adds nodes but no edges beyond

the cohort parquet and reproduces the cohort edge counts exactly. The pool stages therefore measure how runtime responds to added isolated or weakly connected nodes at fixed edge counts; they do not scale the edge count that drives per-iteration cost. Per-stage $|V|$ and $|E|$ for both methods are reported in every scaling table so that this limitation is visible, and the scaling claim in Chapter 4 is confined to what those columns support.

3.12.2 Damping sensitivity

PageRank damping is swept over $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort, holding graphs and proxies fixed. The default $d = 0.85$ matches the classical web PageRank setting^[15] and the AWP baseline^[3]. Lower d increases teleportation and flattens scores; higher d emphasizes multi-hop structure and can slow convergence. The sweep tests whether allowance- and transfer-family alignment are artifacts of a single teleportation weight.

3.12.3 Top-token subgraph

A restricted subgraph retains only the top-20 ERC-20 tokens by combined transfer and allowance volume within the observation window. EndorseRank and AWP are recomputed on this subgraph to assess sensitivity to long-tail tokens and possible spam assets. If family-level conclusions reverse when low-volume tokens are removed, the primary result would be fragile to token composition; stability supports interpreting the full-token graphs as representative of major economic activity.

3.12.4 Sample-definition sensitivity

Alignment statistics are recomputed on staged subsamples drawn from the expanded wallet pool using the same deterministic seed. The primary sample for every alignment claim is the matched wallet cohort ($n = 5,521$), whose coefficients are reported with confidence intervals. The pool subsamples differ from the cohort in size and in composition: they admit wallets with connectivity in only one of the two graphs, for which the missing-graph score and proxy are both zero after alignment, and such tied zeros raise rank agreement mechanically. The staged coefficients are

therefore reported as a sensitivity analysis of the sample definition, not as evidence of stability; the question they answer is whether EndorseRank and AWP preserve their relative ordering on the transfer and allowance families when the sample definition changes.

3.13 Reproducibility

All empirical tables in Chapter 4 are produced by an open evaluation pipeline. After Phase 1 parquet artifacts are present, a single evaluation driver (`scripts/run_dissertation_eval.py--real--robustness--benchmark--repeats5`) recomputes checks, alignment statistics with bootstrap intervals, and benchmarks from those inputs and writes one machine-readable summary (`data/processed/eval_summary.json`); a separate exporter (`scripts/export_latex_results.py`) renders every table in Chapter 4 and Appendix 8.1 from that file, so no number is transcribed by hand. Command-line flags select cached BigQuery-derived inputs versus synthetic fixtures and enable damping, top-token, and sample-definition sweeps. Observation window, damping, decay parameters, topic hashes, bootstrap resamples and seed, and benchmark seeds are collected in a single versioned configuration file (`config/margin_config.yaml`). Fixture mode supports offline verification without cloud credentials.

Reproducible elements include: deterministic wallet subsampling via SHA256 seeds; fixed PageRank tolerance and iteration caps; a fixed bootstrap seed and shared resample index matrix; explicit proxy formulas in Section 3.9; the timing protocol of Section 3.11 with per-run timings retained; and logged extraction manifests. Peer reviewers can regenerate alignment and benchmark tables from the same parquet without re-querying BigQuery.

The evaluation source (`A-Skill-Programs/margin_rank`), versioned configuration, and machine-readable result summary are in the public repository https://github.com/abitocodes/phd_works at commit `0c70518a35540c8b56be66b52452cb94ca3fb916`; the commit hash is updated at each submission. An anonymised archive can be supplied on request.

3.14 Sampling considerations and missingness

The matched cohort is a purposeful sample, not a random sample of all Arbitrum addresses. Eligibility requires non-zero connectivity in the approval or transfer reputation subgraph, so results speak to *wallets with observable endorsement- or transfer-graph structure*. Idle wallets, pure holders, and addresses that interact only through non-ERC-20 mechanisms are out of scope by design.

Missingness arises in several forms. Tokens with unknown decimals may be dropped or defaulted during preprocessing; such decisions are logged in the pipeline. Approvals that never emit standard ERC-20 logs (non-standard tokens) are invisible to EndorseRank. Transfers that occur off-chain or through custodial systems never enter AWP. These missingness patterns bias both methods toward on-chain, standard-token activity—acceptable for an on-chain reputation study, but relevant when generalizing to the full user population of a protocol.

Supplemental addresses in the expanded wallet pool ($N = 99,080$) increase node coverage for runtime tests. Alignment claims therefore remain anchored to the matched cohort ($n = 5,521$), while node-count sensitivity uses the pool.

3.15 Implementation notes

PageRank is implemented in Python with a SciPy sparse transition operator and NumPy power iteration. Edge lists are built once per configuration; timed benchmarks exclude I/O and preprocessing and cover operator assembly plus iteration (Section 3.11). Randomness enters only through optional fixture generation; real-data evaluation is deterministic given parquet inputs and configuration seeds. Floating-point differences across machines are expected to be negligible relative to the reported τ precision (three decimal places).

Figures are generated from the same evaluation artifacts as the tables, so reported numerics and plots remain consistent.

3.16 Ethics

This study uses exclusively publicly available blockchain data and does not involve human participants. Institutional ethics review (Unisa SoC/CAES) was completed before empirical data collection. The ethics clearance certificate, reference [Unisa SoC/CAES ethics clearance ref: TBD] dated [TBD], is reproduced in Appendix 8.6.1.

- Privacy and anonymity: Arbitrum One addresses are pseudonymous identifiers, not inherently linked to real-world identities [2]. Pseudonymity is not anonymity: determined adversaries can sometimes link addresses to persons through exchange deposits, public profiles, or clustering heuristics. This study deliberately refrains from such linkage. Collection is restricted to on-chain events—Approval and Transfer—with no scraping or integration of off-chain personal data such as IP addresses, exchange KYC records, social-media handles, or deanonymization graphs. No attempt was made to cluster addresses into real-world entities, to publish address-level rankings that could reasonably identify individuals, or to combine wallet histories with external identity providers. Reported results are cohort-level aggregates (correlations, runtimes, family means), not leaderboards of named users. Where examples are discussed in prose, they refer to abstract roles (e.g., “high-allowance spenders”) rather than concrete addresses.
- Data minimization: Only fields required for graph construction and checks are retained in processed tables (addresses, token amounts, timestamps). Raw logs are stored for auditability of the research pipeline but are not redistributed as a deanonymization aid. Query filters are wallet-scoped where possible to avoid indiscriminate full-chain dumps beyond the study design.
- Data usage and compliance: All queries targeted public datasets under `bigquery-public-data.google.blockchain.arbitrum_one_us` in accordance with Google Cloud terms of service and rate limits. Dry-run byte estimates and per-query caps limit unnecessary scanning. No transactions were broadcast, no smart contracts were called in ways that could disrupt

network operations, and no privileged or non-public indexer access was used.

- **Fairness and transparency:** Graph reputation systems can systematically undervalue new entrants, low-volume participants, or wallets that avoid public approvals [2]. EndorseRank and AWP were therefore audited for such structural biases: allowance graphs favor addresses that receive spending authority; transfer graphs favor addresses embedded in value-flow networks. Limitations—including cold-start effects and Sybil susceptibility of approval graphs—are discussed in Chapter 5. All code paths, parameter choices, and evaluation metrics are documented in the open pipeline (Appendix 8.1) to enable peer review and independent replication.
- **Responsible disclosure and impact:** Although EndorseRank may inform risk assessment in DeFi contexts, it is not financial advice, credit advice, or a consumer reporting product. Human oversight remains necessary for lending, trading, and compliance decisions. Potential adversarial manipulation (for example, Sybil attacks on G_{approve} or wash transfers on G_{transfer}) is acknowledged and discussed in Chapter 5. The study reports aggregated cohort-level findings rather than wallet-level rankings intended for targeting individual users. Any future operational deployment would require additional governance, appeal mechanisms, and fairness monitoring beyond the scope of this research.
- **Summary of methodological commitments:** The design fixes the chain, window, cohort rules, graph formulas, proxy definitions, bootstrap protocol, timing protocol, and statistical tests before interpreting results. Primary claims compare only EndorseRank and AWP on the matched wallet cohort and the spender holdout; the expanded pool and sensitivity sweeps test efficiency and the dependence of results on parameters and sample definition.

Chapter 4 reports the empirical outcomes of this protocol as an Arbitrum One case study: it first characterises the dataset and cohort, then presents the runtime benchmark and its node-count sensitivity, the sensitivity sweeps, the alignment tables with intervals and contrasts, the rank-divergence test, and the temporal holdout, and closes by stating Claims C1–C4.

4 Implementation and Empirical Results

Chapter 3 specified how EndorseRank and the Adaptive Weighted PageRank (AWP) baseline are built, which construct checks are applied, how uncertainty is quantified and how runtime is timed. This chapter applies that design to Arbitrum One and reports what was observed. It is organised as a single case study: one chain, one observation window, one matched cohort for all same-window comparisons, one spender cohort for the temporal holdout and one expanded address pool for node-count sensitivity. Every table in the chapter is rendered by the pipeline’s exporter from the machine-readable summary described in Section 3.13; where a number is discussed in the text it is stated once, with its table as the reference, and confidence intervals accompany every Kendall τ coefficient. The chapter proceeds from the case-study setting (Section 4.1), through the computational benchmark (Section 4.2), the same-window alignment results with difference tests (Sections 4.3–4.4), inter-method rank divergence (Section 4.5), robustness and sensitivity checks (Section 4.6) and the temporal holdout (Section 4.7), to a consolidated statement of the empirical claims C1–C4 (Section 4.8).

4.1 Case study setting: Arbitrum One

Arbitrum One was selected as the case because it is a high-activity Ethereum rollup with mature DeFi protocols (Section 2.2), its logs are available in the public BigQuery export used by the pipeline, and a margin-trading protocol on it (GMX V2) supplies the wallet seed set described below. The observation window is 1 December 2025 to 31 May 2026. Six months is long enough to span several market regimes and to give tenure and active-month proxies room to vary, and short enough that a latest-allowance snapshot remains contemporaneous with the transfer history that AWP aggregates.

4.1.1 Cohorts

Three address sets are used, each for one purpose.

- Matched cohort ($n = 5,521$). Wallets seeded from GMX V2 margin-trading activity that have non-zero connectivity in both the latest-allowance endorsement graph G_{approve} and the time-decayed transfer graph G_{transfer} over the window. All same-window alignment, difference-test and benchmark results refer to this cohort. Requiring connectivity in both graphs ensures that neither method is evaluated on wallets for which it can only return the teleportation floor.
- Holdout spender cohort ($n = 1,335$). Wallets that appear as spenders in at least one allowance during the scoring period (events up to 28 February 2026). Their scores are tested against owner–spender approval pairs first observed between 1 March and 31 May 2026 (Section 3.7).
- Expanded wallet pool ($N = 99,080$). The matched cohort unioned with 93,559 supplemental addresses sampled from monthly ERC-20 activity on the chain. The pool is used only for node-count sensitivity of runtime and for the sample-definition sensitivity analysis (Section 3.12.1); its limitation—supplemental addresses contribute nodes but no additional extracted edges—is restated where the results are reported.

Graph sizes for the matched cohort are given with the benchmark in Table 4.1. The endorsement graph has roughly one ninth as many edges as the transfer graph on the same wallets. That asymmetry is a property of the two representations rather than of the extraction: both graphs are built from the same wallet seed set and window, but EndorseRank keeps only the latest decimal-adjusted allowance per owner–spender pair, so repeated approvals collapse to one directed edge, whereas AWP accumulates every transfer between two wallets into a decayed weight. Differences in scores and alignment are therefore attributable to ranking logic and graph semantics, not to inconsistent sampling.

4.1.2 Implementation

Both methods run through the same Python PageRank solver with identical damping ($d = 0.85$), tolerance (10^{-8}) and iteration cap; only the edge lists differ (Section 3.15). Scores, per-proxy statistics, bootstrap intervals and timings are written to one summary file per run; the tables below were produced from the run of 12 September 2026 whose configuration is recorded in `config/margin_config.yaml` at the commit named in Section 3.13.

4.2 Computational benchmark (Claim C1)

Claim C1 asserts that EndorseRank is computationally cheaper than AWP on identical inputs. Table 4.1 reports PageRank solve time, its standard deviation over the five timed repeats, peak memory during the warm-up solve, iterations to convergence and the size of each method’s evaluation subgraph on the matched cohort, following the timing protocol of Section 3.11.

Table 4.1: Computational benchmark on the matched wallet cohort ($n = 5,521$; AWP/EndorseRank runtime ratio $7.9\times$). Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs. $|V|, |E|$ = nodes and edges of each method’s evaluation subgraph.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$ V $	$ E $
EndorseRank	1.426	0.277	4.6	37.0	6,442	14,727
AWP	11.253	1.541	38.2	100.0	30,791	137,087

Three mechanisms account for the runtime ratio in the table. First, both parts of the timed solve— assembling the sparse transition operator from the edge list and each power-iteration step—are linear in $|E|$ (Appendix 8.3.3), and the endorsement graph has about one ninth of the transfer graph’s edges. Second, EndorseRank converges in roughly a third of the iterations AWP needs; the transfer operator, with its long-tailed decayed weights, mixes more slowly. Third, peak memory follows $|E|$ as well, since adjacency, weight and residual buffers scale with the edge list. The standard deviations show that run-to-run noise is an order of magnitude smaller than the gap between methods.

Two remarks bound the claim. The benchmark times the solver call on pre-built edge lists; end-to-end refresh cost includes extraction, decoding and edge-list construction, which are shared or partially shared across methods and are not part of C1. And “faster” is not “better”: the runtime result acquires meaning only when paired with the alignment results of Sections 4.3–4.4, which is why Claim C4 combines the two.

4.2.1 Node-count sensitivity on the expanded pool

Table 4.2 reports runtime at three staged node counts drawn from the expanded pool by deterministic SHA256 subsampling, together with the per-stage $|V|$ and $|E|$ of each method’s subgraph. In-cohort stages ($n \leq 5,521$) are given in Appendix 8.1 (Table 8.1).

Table 4.2: Runtime versus evaluation-set size on the expanded wallet pool ($N = 99,080$; SHA256 subsampling seed `benchmark-tier2-v1`). Each stage keeps the edges that touch at least one sampled wallet, so $|V|$ and $|E|$ are reported per stage; the pool adds addresses but no additional extracted edges beyond the matched-cohort subgraph, hence the full-pool row reproduces the matched-cohort graph. Speedup = AWP/ER runtime ratio within the same row. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
10,000	0.130	1.365	10.5×	1,296	2,055	8,489	26,732
50,000	0.752	6.638	8.8×	4,440	9,190	19,990	92,887
99,080	1.426	11.253	7.9×	6,442	14,727	30,791	137,087

Two features of the table need to be read together with Section 3.12.1. First, the full-pool row is numerically identical to the matched-cohort row of Table 4.1. This is not a copy: the supplemental addresses have no extracted approval or transfer logs, so the induced subgraph at $N = 99,080$ contains exactly the cohort edges, and the timing protocol times each distinct graph once per session and reuses that measurement wherever the same graph is tabulated. The same measurement therefore appears in the $d = 0.85$ row of Table 4.8. Second, $|E|$ is not monotone in n across stages, because each stage is the subgraph induced by whichever wallets the hash order selects; the $n = 10,000$ stage happens to select comparatively few cohort wallets and hence few edges. What the table supports is a bounded statement: across stages whose edge counts vary by a factor of

five to seven, the AWP/EndorseRank runtime ratio stays within roughly $8\text{--}10\times$ and EndorseRank remains in the sub-two-second regime. It does not show how either method scales when the edge count itself grows with the address population, since the second-tier extraction that would have supplied those edges did not complete.

Figure 4.1 plots the in-cohort and pool stages on a common log axis.

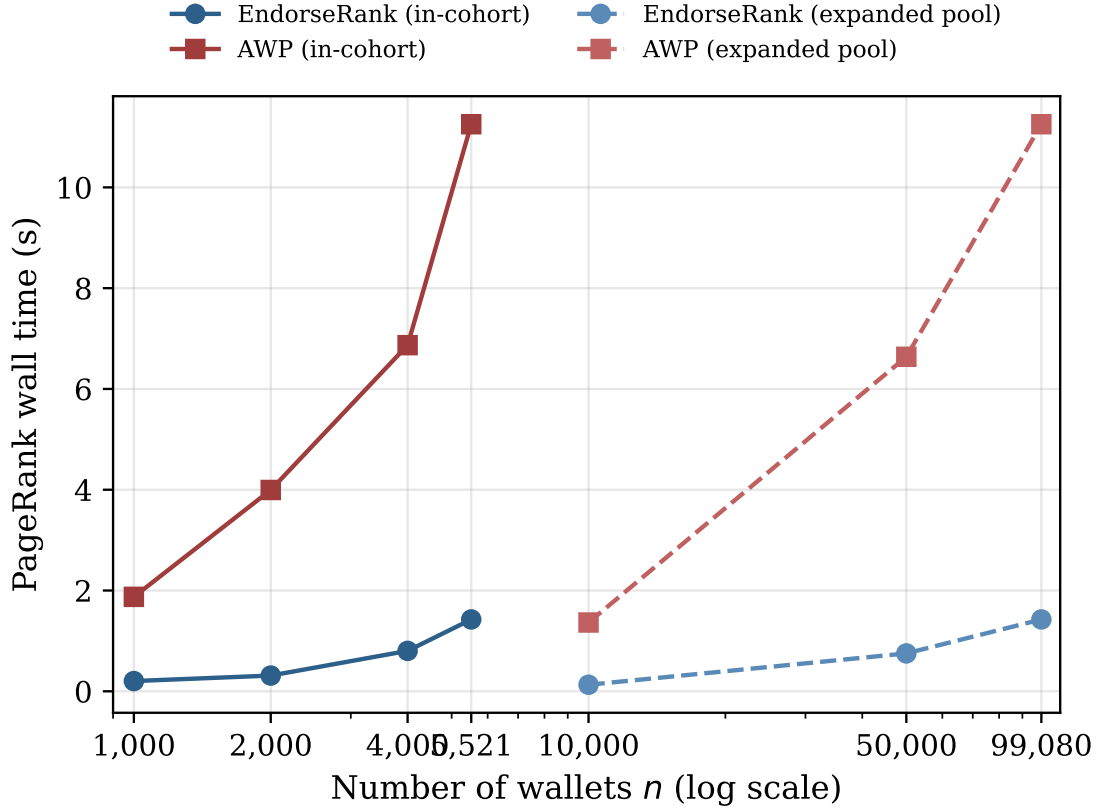


Figure 4.1: PageRank wall-clock runtime versus number of sampled wallets (log x -axis). Solid lines: in-cohort stages ($n \leq 5,521$, Table 8.1). Dashed lines: expanded-pool stages ($N = 99,080$, Table 4.2). Pool stages add addresses without adding extracted edges, so the rightmost pool point coincides with the full matched cohort.

4.3 Same-window alignment: EndorseRank vs. AWP (Claims C2–C3)

Claims C2 and C3 ask whether each method recovers its intended construct on contemporaneous local statistics and whether the winners are construct-specific. Table 4.3 reports family-mean Kendall τ with 95% paired-bootstrap intervals for the transfer, allowance and Sybil-stability families, together with the inter-method agreement between the two score vectors. Figure 4.2 shows the same family means as a heatmap.

Table 4.3: Family-mean Kendall τ with bootstrap confidence intervals on the matched cohort ($n = 5,521$). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Family	ER mean τ	ER 95% CI	AWP mean τ	AWP 95% CI
Transfer	0.333	[0.314, 0.353]	0.534	[0.522, 0.547]
Allowance	0.355	[0.322, 0.384]	-0.030	[-0.045, -0.017]
Sybil	0.233	[0.214, 0.253]	0.286	[0.274, 0.300]
EndorseRank vs. AWP score agreement: $\tau = 0.316$, CI [0.298, 0.336]				

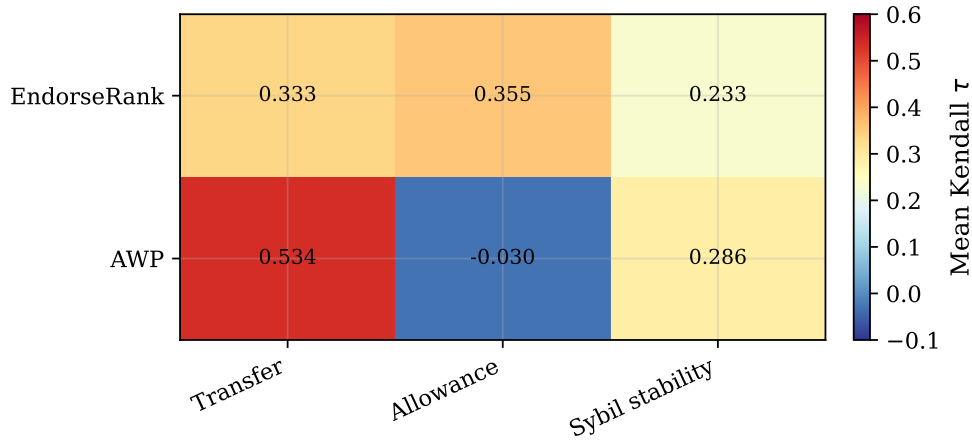


Figure 4.2: Family-mean Kendall τ on the matched cohort ($n = 5,521$) for EndorseRank and AWP; the values are those of Table 4.3.

The pattern is crossed. AWP aligns most strongly with the transfer family and is slightly negative on the allowance family; EndorseRank aligns most strongly with the allowance family and moderately with the transfer family. Whether each contrast exceeds sampling variation is tested in Table 4.4,

which reports the six pre-specified paired differences of Section 3.10.

Table 4.4: Paired bootstrap tests of differences between Kendall τ coefficients (matched cohort, $n = 5,521$). Family entries use the family-mean τ ; the same wallet resamples are used for a and b , so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.355	0.333	+0.022	[-0.026, 0.067]	0.810
AWP transfer minus EndorseRank transfer	0.534	0.333	+0.201	[0.182, 0.218]	1.000
EndorseRank allowance minus AWP allowance	0.355	-0.030	+0.385	[0.350, 0.421]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.286	0.233	+0.052	[0.038, 0.065]	1.000
AWP GMX-success minus EndorseRank GMX-success	0.179	0.096	+0.083	[0.068, 0.100]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.356	0.344	+0.012	[-0.041, 0.060]	0.675

Four of the six contrasts have intervals that exclude zero: AWP’s transfer alignment exceeds EndorseRank’s, EndorseRank’s allowance alignment exceeds AWP’s, and AWP exceeds EndorseRank on the Sybil-stability and (archived) GMX-success families. Two do not. EndorseRank’s allowance alignment is not statistically higher than its own transfer alignment (contrast i), and EndorseRank’s in-approve-degree alignment is not statistically higher than its in-degree alignment (contrast vi). These two null results bound the interpretation of the allowance family. EndorseRank is a global smoothing of spender in-approve degree, so its agreement with allowance proxies is an intended-construct check, not an external test. The between-method statement that follows is licensed: transfer centrality (AWP) carries no information about who receives spending authorisation, whereas endorsement centrality (EndorseRank) does, and the reverse holds for transfer volume. The within-method statement that EndorseRank “prefers” allowance over transfer signal is not licensed.

Claim C2 is supported in the sense that each method achieves non-trivial alignment on its own construct with an interval well away from zero. Claim C3 is supported by the two between-method contrasts (ii and iii) whose intervals exclude zero in opposite directions: neither method dominates both checks.

4.4 Proxy family detail

The family means summarise per-proxy coefficients. This section reports Spearman ρ , Kendall τ and the 95% interval of τ for each proxy. Proxies are oriented so that higher values imply higher expected reputation before correlation, and the transfer proxies are those used to validate AWP in its source paper^[3].

4.4.1 Transfer family

Table 4.5: Domain alignment with transfer proxies (inbound in-degree and in-value, the validation proxies used for the AWP baseline). Kendall τ and Spearman ρ between reputation scores and proxy rankings. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-degree	0.422	0.344	[0.324, 0.366]
In-value (sum inbound)	0.398	0.321	[0.303, 0.342]
<i>AWP</i>			
In-degree	0.791	0.617	[0.604, 0.631]
In-value (sum inbound)	0.610	0.450	[0.436, 0.467]

AWP’s agreement with in-degree is the largest coefficient in the chapter, and its agreement with in-value is smaller but still far above EndorseRank’s. Both are primary-construct checks for AWP, since it ranks on the time-decayed transfer graph; the informative observation is that AWP does not collapse to in-degree—logistic decay and value weighting move a measurable share of pairwise orderings. EndorseRank’s transfer coefficients are moderate and their intervals overlap neither zero nor AWP’s intervals. Wallets that receive many transfers are somewhat more likely to be endorsed spenders, but the two behaviours are far from interchangeable. If a protocol’s notion of reputation is inbound economic flow, AWP is the appropriate method on this evidence.

Table 4.6: Domain alignment with allowance proxies (spender-side in-approve degree and value). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-approve degree	0.360	0.356	[0.323, 0.385]
In-approve value (sum)	0.360	0.354	[0.321, 0.383]
<i>AWP</i>			
In-approve degree	-0.037	-0.030	[-0.045, -0.017]
In-approve value (sum)	-0.037	-0.030	[-0.045, -0.017]

4.4.2 Allowance family

EndorseRank’s coefficients on in-approve degree and in-approve value are nearly equal, which indicates that on this cohort being approved by many owners and being approved for large aggregate value convey similar ranking information under latest-allowance semantics. AWP’s coefficients are slightly negative with intervals that exclude zero: transfer centrality does not recover authorisation. That dissociation is the empirical core of the same-window contribution. If transfer-based reputation were a sufficient proxy for endorsement, AWP would show at least moderate allowance alignment; it does not, so protocols that care about which addresses are trusted to move tokens on behalf of others cannot substitute AWP for an allowance-aware method. The caveat of Section 4.3 applies: EndorseRank’s own allowance coefficients are a construct check, and the out-of-window test of the same construct is the holdout of Section 4.7.

4.4.3 Sybil-adjusted stability family

AWP exceeds EndorseRank on tenure and active months, as expected from time-decayed transfer edges that reward sustained activity; the paired difference on the family mean (Table 4.4, contrast iv) excludes zero. The inbound-counterparty ratio is weak for both methods and negative for AWP. This family is a robustness note on how much longitudinal transfer information each method carries, not a headline claim.

Table 4.7: Domain alignment with Sybil-adjusted stability proxies from transfer activity. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
Inbound counterparty ratio	0.093	0.073	[0.046, 0.097]
Transfer tenure (days)	0.366	0.297	[0.276, 0.318]
Active months	0.381	0.330	[0.310, 0.351]
<i>AWP</i>			
Inbound counterparty ratio	-0.063	-0.061	[-0.080, -0.040]
Transfer tenure (days)	0.602	0.436	[0.421, 0.452]
Active months	0.622	0.482	[0.466, 0.499]

4.5 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 ask whether EndorseRank and AWP order the same wallets differently. The inter-method Kendall τ between the two score vectors, with its interval, is given in the last row of Table 4.3. The rankings are related but far from interchangeable: agreement exceeds chance because economically active wallets participate in both approval and transfer markets, yet a large fraction of pairwise orderings disagree.

Figure 4.3 shows the marginal score distributions. Both are right-skewed, as PageRank on sparse directed graphs typically is, but they differ in shape and in the number of wallets that sit at the teleportation floor. Skewness alone does not establish divergence, so Figure 4.4 plots the ranks directly: the cloud is positively sloped but wide, with substantial off-diagonal mass.

Rank divergence is the structural counterpart of construct-specific alignment. If the two methods produced nearly identical orderings, the family-level winners of Table 4.3 could be noise around a shared ranking; the observed divergence, together with the opposite-signed between-method contrasts of Table 4.4, supports reading the two as related but distinct instruments. H1 is supported. Operationally, a protocol that currently uses transfer-based scores cannot assume that switching to, or blending in, allowance-based scores will preserve its wallet ordering; the largest disagreements are expected among transfer hubs that are rarely endorsed as spenders and among endorsed

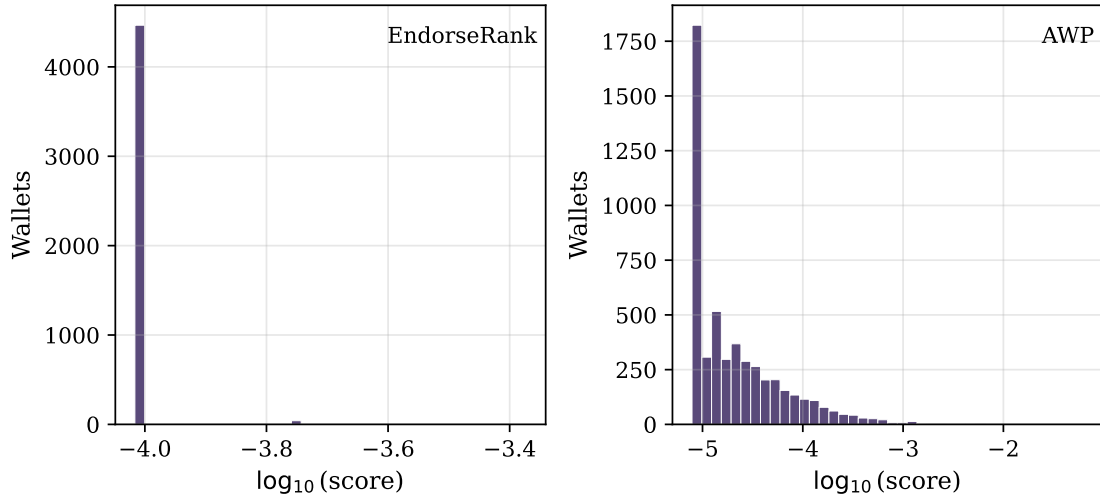


Figure 4.3: Marginal distributions of EndorseRank and AWP scores on the matched cohort ($n = 5,521$), on a $\log_{10}$ scale for wallets with positive score (EndorseRank has 966 zero-score wallets and AWP 381, omitted from the histograms).

spenders with small transfer footprints.

4.6 Robustness and sensitivity checks

The results above would be of limited use if they depended on a single damping value, a single token mix or a single sample definition. Three checks follow. The first two are robustness checks in the strict sense: the sample is held fixed and a parameter or the token universe is perturbed. The third is a sensitivity analysis: the sample definition itself is changed, and the coefficients are expected to move.

4.6.1 Damping

Family means are flat across $d \in \{0.75, 0.85, 0.95\}$ to the third decimal for EndorseRank and move by at most a few thousandths for AWP; the crossed pattern is not an artefact of the conventional $d = 0.85$. Because allowance edges encode a snapshot rather than a long path structure, changing how much mass teleports does not reorder wallets enough to alter family-level alignment. Runtimes do vary with d . AWP is markedly slowest at $d = 0.95$, consistent with the slower mixing of a high-

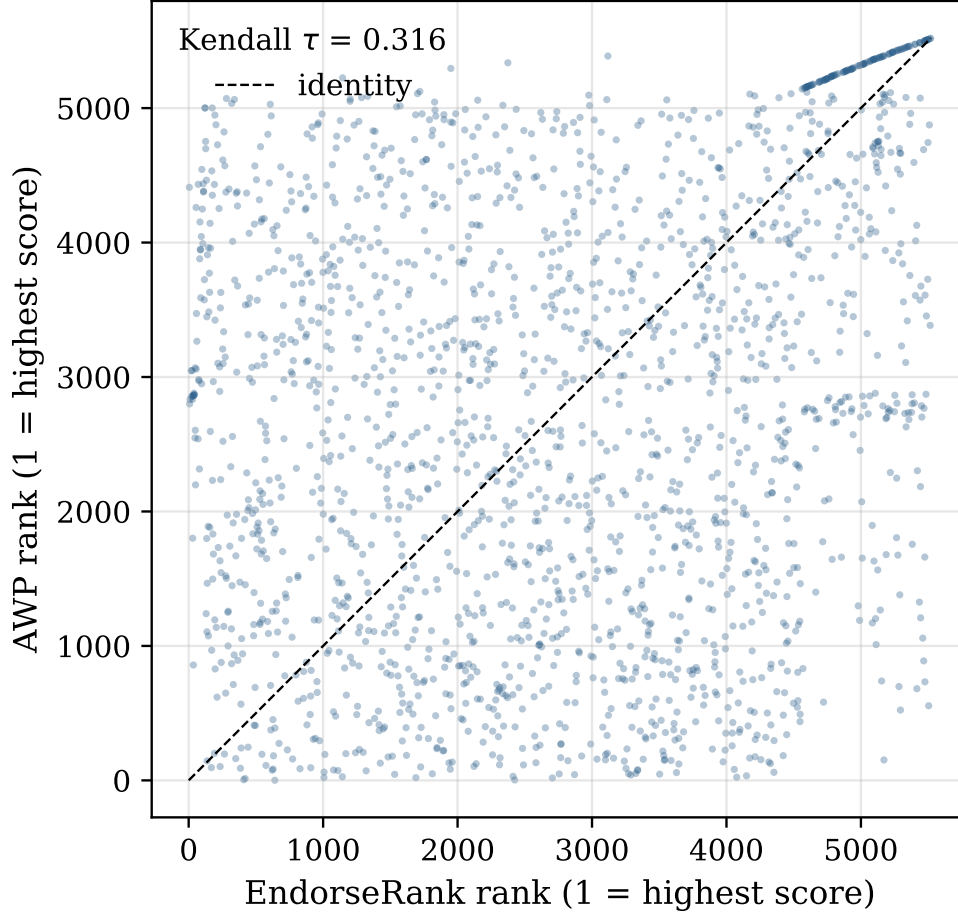


Figure 4.4: Rank–rank scatter of EndorseRank versus AWP on the matched cohort (random sub-sample of 2,000 wallets for legibility). Points on the dashed identity line would indicate identical orderings.

Table 4.8: Robustness: mean Kendall τ under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched alignment cohort ($n = 5,521$). The $d = 0.85$ runtimes are the same measurements as Table 4.1 (one timing session; each graph–damping pair is timed once).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)
0.75	0.355	-0.031	0.333	0.531	2.276	13.587
0.85	0.355	-0.030	0.333	0.534	1.426	11.253
0.95	0.355	-0.030	0.333	0.536	0.954	18.649

damping operator; EndorseRank’s variation across the sweep is of the same order as its between-run spread in Table 4.1 and is not interpreted. The runtime ratio therefore depends on the damping setting even though the alignment conclusion does not. Figure 4.5 plots the family means.

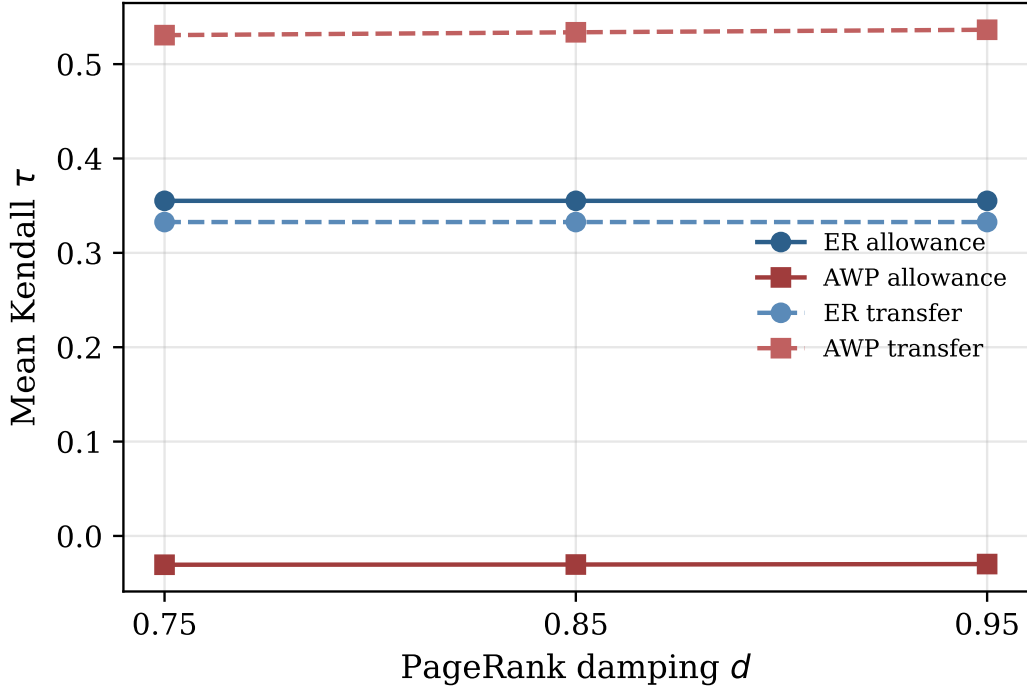


Figure 4.5: Damping sensitivity of family-mean Kendall τ for the allowance and transfer families, $d \in \{0.75, 0.85, 0.95\}$ (values of Table 4.8).

4.6.2 Top-token subgraph

Table 4.9: Robustness: mean Kendall τ on the top-20 ERC-20 token subgraph (matched cohort), next to the full-token value of Table 4.3 for the same family.

Method	Transfer		Allowance		Sybil	
	top-20	full	top-20	full	top-20	full
EndorseRank	0.331	0.333	0.355	0.355	0.234	0.233
AWP	0.532	0.534	-0.031	-0.030	0.283	0.286

Restricting both graphs to the twenty highest-volume ERC-20 tokens changes every family mean by at most a few thousandths. The alignment pattern is therefore not driven by long-tail tokens with

sparse or spam-like edges; the economically salient core of ERC-20 activity suffices to reproduce the EndorseRank–AWP contrast.

4.6.3 Sample-definition sensitivity

Table 4.10: Sensitivity of alignment to the evaluation set: mean Kendall τ and runtime at staged subsample sizes drawn from the expanded wallet pool ($N = 99,080$; seed `benchmark-tier2-v1`). Rows are evaluated on different wallet sets from the matched cohort, so τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. Runtimes are the same measurements as Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)	Speedup
10,000	0.343	0.242	0.821	0.904	0.130	1.365	10.50
50,000	0.330	0.232	0.828	0.915	0.752	6.638	8.83
99,080	0.327	0.230	0.828	0.916	1.426	11.253	7.89

Table 4.10 recomputes the transfer and allowance family means on the staged pool subsamples used for the runtime scaling. The coefficients are much larger than on the matched cohort, and AWP’s allowance coefficient turns positive. Neither change is evidence about the methods. Pool subsamples admit wallets that have no edges in one or both graphs; for such wallets the score and the proxy are both zero after alignment, and tied zeros at the bottom of both rankings raise Kendall τ mechanically (Section 3.12.4). The table is therefore reported as a sensitivity analysis of the sample definition, and it answers a narrower question: does the relative ordering of the two methods on each family survive a change of population? It does—AWP stays ahead on transfer and EndorseRank on allowance at every stage—but the absolute values in the table must not be compared with those of Table 4.3, whose cohort excludes exactly the wallets that inflate them.

The damping and top-token checks support reading Claims C1–C3 as stable under parameter and token-mix perturbation. The sample-definition check shows that the comparative structure, not the decimal values, is what travels across populations.

4.7 Temporal holdout of future new approvals (RQ4)

The same-window checks cannot separate a method from its own construct. RQ4 therefore asks whether endorsement centrality measured up to a cut-off predicts endorsement behaviour after it. Scores are frozen at 28 February 2026 and labels are new owner–spender approval pairs first observed from 1 March to 31 May 2026, counted on the spender; the protocol is that of Section 3.7. Table 4.11 reports the result.

Table 4.11: Temporal holdout on the spender cohort ($n = 1,335$): Kendall τ between scores frozen at 28 February 2026 and the number of new owner–spender approval pairs received between 1 March and 31 May 2026. CI = 95% percentile bootstrap over wallets (200 resamples). 222 spenders received at least one new pair.

Method	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank	0.522	0.479	[0.426, 0.529]
AWP	0.053	0.044	[−0.011, 0.095]

EndorseRank’s holdout coefficient is larger than its same-window allowance coefficient and its interval excludes AWP’s, whose interval includes zero. Because the labels are built from events that post-date every scored event, this is out-of-window evidence for the endorsement construct, and it is the strongest single result in the chapter for the claim that EndorseRank measures something about spenders that transfer centrality does not. Two limits apply. The label is the same construct in the future tense—new approvals, not loss avoidance or repayment—so the result says nothing about credit risk. And the label is sparse: most spenders received no new pair in the outcome window, so the coefficient is driven by the ordering among the minority that did.

Other candidate labels computed for the holdout are not reported as support. Revoke counts and the drain heuristic correlated positively with EndorseRank, which has the wrong sign for an endorsement reading (a heavily endorsed spender is also a frequently revoked one), and Aave liquidation events on this spender set were empty. The trading-success comparison of the research proposal was withdrawn for the reason given in Section 2.8: the outcome-native reference it relied on is built from the same GMX events as the proxies it would be validated against.

4.8 Primary claims (C1–C4)

The results of this chapter support the following claims, mapped to hypotheses and research questions in Table 1.2.

- C1. On identical wallet sets EndorseRank solves PageRank in a fraction of AWP’s wall-clock time, with proportionally lower memory and fewer iterations, because its latest-allowance graph has about one ninth of the transfer graph’s edges (Table 4.1). The ratio is stable across in-cohort and pool stages whose edge counts vary by a factor of five to seven (Tables 4.2 and 8.1); the pool stages do not test growth in edge count beyond the cohort graph.
- C2. Each method recovers its intended construct on same-window local statistics, with bootstrap intervals well away from zero (Table 4.3). These are construct checks, not external validation.
- C3. The winners are construct-specific: AWP exceeds EndorseRank on the transfer family and EndorseRank exceeds AWP on the allowance family, with paired-difference intervals that exclude zero in opposite directions (Table 4.4); the two score vectors agree only moderately (Table 4.3). EndorseRank’s allowance lead over its own transfer alignment is not statistically established.
- C4. The efficiency and construct results combine into a trade-off. EndorseRank is the cheaper method and the only one that aligns with authorisation checks, and on the spender holdout it predicts future new approvals where AWP does not (Table 4.11). AWP remains the stronger method for transfer-hub detection and longitudinal stability.

RQ1 is answered by the inter-method agreement and Figure 4.4; RQ2 by the family tables and difference tests; RQ3 by the benchmark tables; RQ4 by the holdout. The claims do not retire AWP. They report a construct-aware comparison with quantified uncertainty: EndorseRank is cheaper, aligns with allowance checks, diverges moderately from AWP and predicts later new approvals on the spender holdout.

Chapter 5 interprets these findings against the research questions and objectives of Chapter 1, draws

out their theoretical, practical and societal implications, and examines the threats to validity and limitations that bound them.

5 Discussion

Chapter 4 reported what was measured on Arbitrum One. This chapter asks what those measurements mean. It first answers each research question and judges each objective of Chapter 1 against the criterion stated there (Sections 5.1–5.2). It then draws out theoretical implications (Section 5.3), practical deployment implications (Section 5.4) and the societal significance of a cheap, auditable, authorisation-aware reputation signal (Section 5.5), formalises the cost of a Sybil attack on an allowance graph (Section 5.6), and closes with regulatory and ethical considerations, threats to validity and limitations (Sections 5.7–5.9). Numerical results are not restated here; each interpretation names the table of Chapter 4 it rests on. Same-window statements refer to the matched cohort and holdout statements to the spender cohort unless noted.

5.1 Findings by research question

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP produce meaningfully different wallet orderings on an identical sample. The inter-method Kendall τ and its interval (Table 4.3, last row) sit well inside the open interval between unrelated and identical orderings: correlation is present, because both methods draw on overlapping wallet activity in one window, but a large share of pairwise orderings disagree (Figure 4.4).

The divergence is structurally expected. The endorsement graph encodes current spending authorisation with about one ninth as many edges as the transfer graph, which encodes time-decayed economic flow^[3]. PageRank propagates influence along whatever edges it is given under the same damping convention^[15], and the sensitivity of the stationary vector to the edge set is well characterised^[23]; different edge sets therefore yield different stationary mass. A wallet that receives large

transfers without granting allowances ranks highly under AWP and modestly under EndorseRank, and conversely for a wallet that authorises spenders without being a transfer hub. H1 is supported. The practical corollary is that the two scores are not substitutes: adopting EndorseRank for efficiency means accepting a different ranking geometry, not a faster approximation of AWP.

5.1.2 RQ2: Construct-specific alignment

RQ2 asks which method aligns more strongly with contemporaneous allowance checks and with contemporaneous transfer checks. Tables 4.3 and 4.4 show construct-specific winners. AWP leads on the transfer family and EndorseRank on the allowance family, and both between-method paired differences exclude zero in opposite directions.

Two qualifications matter for how H2 is read. First, each method’s own-construct agreement is partly mechanical, because the proxies of a family are computed from the same event stream that builds the corresponding graph; RQ2 is therefore a construct-validity observation in the sense of Cronbach and Meehl ^[22], not external validation. Second, the within-method contrasts are null: EndorseRank’s allowance alignment is not statistically higher than its own transfer alignment, and its agreement with in-approve degree is not statistically higher than its agreement with transfer in-degree (Table 4.4, contrasts i and vi). The evidence therefore supports the between-method statement—transfer centrality does not recover authorisation, and endorsement centrality does not recover transfer volume as well as AWP—but not a claim that EndorseRank is “more about” allowances than about transfers within its own ranking. H2 is supported in that construct-specific, between-method form.

5.1.3 RQ3: Computational efficiency

RQ3 asks whether EndorseRank is computationally more efficient than AWP on the same matched sample. Table 4.1 answers affirmatively on wall-clock time, memory and iterations, and reports the edge counts next to the runtimes so that the ratio can be attributed: the endorsement graph is about one ninth the size of the transfer graph, and per-iteration cost is linear in $|E|$ ^[23]. The

advantage follows from representing authorisation as a latest-allowance snapshot rather than as an accumulated, logistically decayed transfer history^[3]; nothing in the solver differs between the two methods.

The node-count sensitivity of Table 4.2 qualifies the claim; it does not extend it. The runtime ratio stays within a narrow band across staged node counts, but the pool stages add addresses without adding extracted edges, so they do not test how either method scales when the edge population itself grows. Damping and top-token perturbations (Tables 4.8 and 4.9) leave the alignment pattern unchanged; the runtime ratio does move with damping. H3 is supported for the matched cohort and for the bounded node-count experiment that was actually run.

5.1.4 RQ4: Temporal holdout of future new approvals

RQ4 asks whether scores frozen at the end of February predict new owner–spender approvals in the following three months on inbound spenders. Table 4.11 shows that EndorseRank does and AWP does not; EndorseRank’s interval excludes zero and lies entirely above AWP’s, whose interval includes zero. Because the labels post-date every scored event, this is the one result in the thesis that is not a same-window construct check, and it is the strongest evidence that endorsement centrality carries information about spenders that transfer centrality does not.

RQ4 differs from the research proposal, which planned a comparison against a trading-success reference. That comparison was withdrawn because the outcome-native reference was built from the same GMX events as the proxies it would have been validated against (Section 2.8); the holdout replaces it with a label that is out of window but still of the same construct—new approvals, not profit or repayment. The limit of RQ4 is therefore that it demonstrates temporal predictive validity of the endorsement construct, not credit-risk validity.

5.2 Achievement of the research objectives

Chapter 1 stated four objectives with a measurement and a criterion each. Table 5.1 judges them against the evidence.

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of inter-method τ excludes 0 and 1	Table 4.3, last row: interval lies strictly inside $(0, 1)$	Met
O2	Each method’s own-construct $\Delta\tau$ against the other method is positive with an interval excluding 0	Table 4.4, contrasts ii (AWP on transfer) and iii (EndorseRank on allowance)	Met
O3	Runtime ratio reported together with the edge ratio under one fixed protocol	Tables 4.1, 4.2 and 8.1: mean, SD, memory, iterations, $ V $, $ E $ per configuration	Met
O4	EndorseRank holdout interval excludes 0 and point estimate exceeds AWP	Table 4.11	Met

Two findings that were not objectives also shape the conclusions. The within-method contrasts of Table 4.4 did not reach significance, which narrows what can be said about EndorseRank’s allowance family (Section 5.1.2), and the expanded-pool experiment turned out to test node count only, not edge growth (Section 5.1.3). Both are reported as limits on the claims. They are not folded into the objectives.

5.3 Theoretical implications

The findings bear on three strands of theory: hyperlink-style propagation on financial graphs, construct validity for on-chain scores, and the economics of trust under pseudonymity.

5.3.1 Propagation semantics beyond transfers

PageRank’s original formulation treats inbound hyperlinks as citations^[15]. Do and Do^[4] carried that metaphor to Ethereum transactions, and Do, Do and Nguyen^[3] refined it into AWP by weighting transfer edges with value and a logistic decay of age. EndorseRank carries the metaphor to ERC-20

allowances: an owner who authorises a spender issues a directed endorsement of spending authority ^[18]. The divergence measured under RQ1 shows that changing the edge semantics changes the ranking geometry even when the node set is held fixed. Graph-based reputation is therefore a family of methods whose validity depends on which relationships are encoded as edges ^[23]. Trust-propagation systems in peer-to-peer and web settings made the same point from the other side: in EigenTrust ^[16] and TrustRank ^[17], seed sets and edge definitions dominate outcomes. EndorseRank contributes an edge definition native to a token standard rather than to a message-passing overlay or a hyperlink graph.

5.3.2 Multi-construct domain alignment

The check design separates transfer centrality from allowance delegation and tests each method on both. The resulting pattern—each method converging with its own construct and diverging from the other’s—is the pattern that classical construct-validity reasoning predicts for indicators that measure different things ^[22], and Messick’s ^[33] unified view adds that the consequences of using a score are part of its validity, which is why the deployment scenarios below are stated in terms of what each score can and cannot be used for. Rank correlations ^{[20],[21]} operationalise the reasoning for ordinal reputation scores without imposing classification thresholds, and the paired bootstrap ^[34] turns the pattern into tested differences rather than eyeballed ones. The null within-method contrasts are a reminder that convergent evidence on a method’s own construct is not the same as discriminant evidence within the method.

5.3.3 Reputation under cheap pseudonyms

Public ledgers permit cheap identity creation ^[5], and Ethereum’s account model makes address creation free of any registration step ^[36]. Reputation systems that reward inbound transfers or allowances are therefore exposed to Sybil strategies in which an attacker manufactures edges among controlled addresses ^[11]. The empirical results do not eliminate that threat; they show that allowance and transfer graphs respond differently to the activity present in the observational sam-

ple. Section 5.6 formalises attack cost on allowance graphs and argues that non-zero allowances to external spenders impose opportunity and revocation costs that transfer wash trading does not.

5.4 Practical implications and deployment scenarios

Protocols and infrastructure providers face a design choice between transfer-hub detection and authorisation-aware reputation. The empirical trade-off supports scenario-dependent adoption. It does not identify a single recommended score.

5.4.1 Transfer-hub and flow monitoring

Protocols that prioritise detection of high-flow wallets—for market-maker identification, routing analysis or transfer-based compliance heuristics—have reason to prefer AWP despite its cost. Its transfer alignment (Table 4.5) makes it the better match when the operational target is economic flow centrality^[3]. Its runtime on the matched cohort (Table 4.1) is acceptable for batch analytics and less attractive for near-real-time refresh.

5.4.2 Authorisation-aware and low-latency refresh

Protocols that emphasise explicit spending authorisation—token vaults, allowance-gated integrations, risk engines that treat approvals as trust acts—have reason to prefer EndorseRank. Its allowance alignment (Table 4.6), its holdout result (Table 4.11) and its sparse graph and short solve time (Table 4.1) support frequent recomputation. Operators accept weaker transfer alignment in exchange for authorisation semantics, out-of-window evidence and speed.

5.4.3 Evaluation design and hybrid use

Hybrid deployments are feasible: EndorseRank for continuous authorisation-aware monitoring and AWP for periodic transfer-hub audits. The open pipeline enables cross-protocol replication on any ERC-20 chain with allowance logs (Appendix 8.1). Integration with live risk oracles remains future

work (Chapter 6), but EndorseRank’s efficiency profile is compatible with refresh cadences that a transfer-graph method may not sustain at equal cost.

5.4.4 Industry adoption scenarios

Both methods supply a cheap, standardised and independently recomputable ordinal signal. Value capture is hypothesised to run through fees, utilisation, monitoring cost and reduced reward leakage, not through reducing collateral. The scenarios below map actors to decision problems consistent with the results.

- DEX operators. Protocols that earn from trading fees must decide which wallets to prioritise for monitoring, fee tiers or feature access while refreshing reputation frequently. EndorseRank fits this loop because its score tracks authorisation activity and can be recomputed in seconds without transfer-history aggregation. Any revenue effect would run through fee volume and lower monitoring cost; fee and monitoring policies are the justified levers.
- Lending protocols. Lending markets seek higher capital utilisation without abandoning collateralisation ^[41]; the 2020 stress episodes show what happens when risk parameters lag ^[9]. Reputation can enter as an auxiliary signal for interest-rate or fee tiers and review cadence on already-collateralised positions, not as a substitute for collateral. EndorseRank is the natural input when the protocol already relies on ERC-20 approvals for deposits and borrows (Section 2.7). The scores are not default predictors; any collateral-differentiation experiment remains future work.
- Wallet and security tooling. Users and integrators must decide whether a spender contract is widely authorised. Displaying a spender’s EndorseRank, or the distribution of inbound endorsement mass, as a trust badge operationalises allowance semantics as explicit delegation ^{[18],[19]}. The badge reports graph position under published edge definitions; it is not a security audit or a guarantee against malicious contracts.
- Liquidity providers and market makers. These actors need transfer hubs and flow structure for

inventory and spread management. AWP is the better method for that target^[3]; the operational channel is lower screening cost, not improved profit and loss.

- Airdrop and governance programmes. Token distributions and governance eligibility are routinely gamed by Sybil clusters ^{[11],[32]}. Programmes can require inbound endorsement diversity or apply the closed-cluster mass bound of Section 5.6 (Equation 5.8) as a filter before rewards are paid. Such filters mitigate closed-cluster inflation; they do not stop open-cluster attacks that recruit honest inbound endorsements.
- Analytics and compliance providers. Exchanges, funds and compliance teams need auditable counterparty rankings that third parties can recompute. The open pipeline supports publishing EndorseRank and AWP under disclosed windows and parameters; auditability is the differentiator relative to opaque machine-learning scores ^[2]. Providers should disclose Sybil exposure and the fact that the scores are not profit forecasts alongside any surfaced rank.

These scenarios do not treat reputation as a substitute for collateral. They treat a standardised, auditable, low-latency ordinal signal as a way to lower the cost of counterparty differentiation.

5.4.5 Standardised scores as shared infrastructure

The practical value of a reputation score, like that of any standardised financial ratio, is that it is public, comparable and cheap to recompute, so that many parties can reason from the same number while knowing how it can be gamed. Forecasting power is secondary. EndorseRank and AWP can play that infrastructural role for pseudonymous wallets: they are computed from public logs by an open pipeline (Appendix 8.1), cheap enough to refresh, and independently verifiable. Neither is a profit forecast, and both are useful only if their construction is public and their failure modes are named (Sections 5.6 and 5.7).

5.5 Societal and practical impact

Counterparty screening in DeFi is expensive and unevenly available. Well-resourced actors run proprietary risk engines over full transfer histories; retail users rely on informal signals or none. A score that can be refreshed at a fraction of the transfer-graph cost (Table 4.1) and that reads directly off the approvals users already grant lowers the resource barrier to systematic screening. Because approvals are also the mechanism through which most token-draining exploits operate (Section 2.17), a spender-centred endorsement score speaks to a risk that users face directly and that transfer volume does not capture.

For protocol operators, the results support a shift from a single “reputation” to construct-specific instruments chosen for the decision at hand. That shift reduces the risk of mis-specified gating—using a flow score to decide an authorisation question or the reverse—and, because both instruments here are computed from public data by a published pipeline, it allows gating rules to be audited by the users they affect. The efficiency result makes continuous monitoring economically feasible for smaller protocols that could not sustain periodic transfer-graph refresh at equal cost.

For regulators and the public interest, the contribution is an existence proof that a useful on-chain reputation signal can be fully transparent: every input event is public, every parameter is versioned and every table is regenerable. That differs from the black-box decentralised credit scores whose due-process risks Packin and Lev-Aretz^[2] describe, and it matches the explainability that regulatory analyses of DeFi call for ^[63]. The same transparency imposes an obligation, developed in Section 5.7, not to let an ordinal graph statistic be mistaken for identity, creditworthiness or proof of personhood.

The evidence is from one chain and one six-month window, and it concerns the endorsement construct, not credit outcomes. Social benefit therefore depends on deployments that respect the construct boundary. Any reputation signal that gates value also invites strategic behaviour. The cost model of the next section shows where that behaviour is expensive; it does not show that the behaviour is impossible.

5.6 Sybil attack cost model on allowance graphs

Sybil attacks exploit cheap pseudonyms to manufacture reputation ^[11]. On transfer graphs, an attacker can wash-trade value among controlled wallets, inflating in-degree and in-value at the cost of fees and inventory risk ^[29]. On allowance graphs the attack surface differs, because edges represent authorisation rather than completed transfers. This section formalises a minimal cost model for inflating EndorseRank through fabricated allowances.

5.6.1 Attack setup

Let $\mathcal{A}$ be the set of addresses controlled by an attacker and $\mathcal{H}$ the set of honest wallets in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) with weight $w_{uv} > 0$ denotes that u currently authorises v to spend decimal-adjusted token units totalling w_{uv} . EndorseRank computes the PageRank vector $\mathbf{r}$ on G_{approve} with damping $d \in (0, 1)$ ^[15]:

$$\mathbf{r} = d P^\top \mathbf{r} + \frac{1-d}{|V|} \mathbf{1}, \quad (5.1)$$

where P is the row-stochastic matrix obtained by normalising outgoing allowance weights, with the standard handling of dangling nodes ^[23].

The attacker seeks to maximise the mean EndorseRank of a target subset $T \subseteq \mathcal{A}$,

$$J(T) = \frac{1}{|T|} \sum_{i \in T} r_i, \quad (5.2)$$

subject to a budget on attack cost defined below.

5.6.2 Cost components

Fabricating an allowance edge (u, v) with weight w_{uv} incurs at least three cost components.

1. Transaction cost. Submitting `approve` or `permit` consumes gas. Let c_{gas} be the expected fee

per allowance update. For m updates,

$$C_{\text{gas}} = m c_{\text{gas}}. \quad (5.3)$$

2. Capital lock and opportunity cost. A non-zero allowance to an external spender $v \notin \mathcal{A}$ exposes tokens owned by u to transfer by v . Even if $v \in \mathcal{A}$, maintaining large allowances to contracts or routers may be required to mimic honest behaviour. Let $\ell(w_{uv})$ be the opportunity cost of capital associated with weight w_{uv} over the reputation refresh horizon (for example, forgone yield). Then

$$C_{\text{cap}} = \sum_{(u,v) \in E_{\mathcal{A}}} \ell(w_{uv}), \quad (5.4)$$

where $E_{\mathcal{A}}$ denotes attacker-created edges.

3. Revocation and detection cost. Honest users revoke stale allowances; attackers who never revoke may be distinguishable. Let c_{rev} be the expected cost of periodic revocation and re-approval needed to maintain a target edge set over k refresh epochs:

$$C_{\text{rev}} = k c_{\text{rev}}. \quad (5.5)$$

Total attack cost is

$$C = C_{\text{gas}} + C_{\text{cap}} + C_{\text{rev}}. \quad (5.6)$$

5.6.3 Influence and a cost-effectiveness bound

Under standard PageRank sensitivity results, the influence of an edge on $\mathbf{r}$ is mediated by the stationary mass at the source and the normalised outgoing weight ^[23]. A coarse upper bound on the gain in $J(T)$ from adding edges among $\mathcal{A}$ alone—a closed Sybil cluster with no honest inbound endorsements—is limited by the teleportation mass:

$$\sum_{i \in \mathcal{A}} r_i \leq |\mathcal{A}| \cdot \frac{1-d}{|V|} + d \cdot \sum_{i \in \mathcal{A}} r_i, \quad (5.7)$$

which rearranges to

$$\sum_{i \in \mathcal{A}} r_i \leq \frac{|\mathcal{A}|}{|V|}. \quad (5.8)$$

A closed attacker cluster therefore cannot capture more than its population share of total rank mass. To exceed that share, the attacker needs inbound endorsements from $\mathcal{H}$, which requires either compromising honest wallets or inducing honest owners to approve attacker-controlled spenders—actions whose cost is typically dominated by C_{cap} and social-engineering effort rather than gas alone.

When the attacker purchases inbound allowances from honest owners (for example, by operating a service that requests approvals), the marginal cost per unit of honest-sourced weight w_{ha} can be written

$$c_{\text{in}}(w_{ha}) = c_{\text{gas}} + \ell_{\text{honest}}(w_{ha}) + \pi(w_{ha}), \quad (5.9)$$

where ℓ_{honest} is compensation paid to the honest owner and π is a risk premium for potential loss from malicious spending. Cost-effectiveness of the attack is then

$$\eta = \frac{\Delta J(T)}{C}, \quad (5.10)$$

with $\Delta J(T)$ the increase in mean target rank attributable to the attack edges.

Equations (5.7)–(5.10) imply that EndorseRank is not Sybil-proof, but that closed-cluster inflation is population-share limited and that cross-cluster inflation requires costly honest-sourced allowances. Transfer wash trading faces different costs (fees and inventory) but does not require exposing spending authority to external addresses. The Sybil-adjusted stability proxies of Chapter 4 partially address observational manipulation; they do not replace adversarial evaluation, which remains future work.

5.7 Regulatory and ethical considerations

On-chain reputation scores occupy a contested space between transparency and fairness. Legal analyses of DeFi emphasise that decentralised markets remain subject to disclosure and consumer-protection concerns even when settlement is on-chain ^[63]. Packin and Lev-Aretz^[2] caution that decentralised credit scoring can reproduce black-box harms if models are opaque, unappealable or coupled to identity systems without due process. EndorseRank and AWP are fully determined by public ledger events and published algorithms; any party can recompute the scores from the open pipeline (Appendix 8.1). That auditability is a partial response to black-box concerns, but it does not resolve distributional effects: wallets with limited approval or transfer history receive low scores regardless of off-chain creditworthiness.

Ethically, scores must not be presented as identity, as creditworthiness in the consumer-lending sense or as proof of unique personhood ^[11]. They are ordinal indicators of on-chain graph position under specified edge semantics. Deployments that gate access to financial services should disclose the graph definition, observation window and known failure modes, including Sybil exposure. Regulatory regimes that require explainability favour graph propagation over opaque machine-learning scores of the kind surveyed by Kotb ^[14], provided operators resist overclaiming predictive power for loan default ^[2].

Privacy is structurally limited on public ledgers ^[5]: reputation computation uses data already broadcast. The ethical obligation is therefore not to avoid analysis but to avoid deanonymisation beyond address-level aggregation and to prevent misuse of scores as persistent biometric-like identifiers. This research analyses addresses only; no off-chain identity linkage is performed (Section 3.16).

5.8 Threats to validity

Interpretation of the empirical claims should account for the following threats, organised by construct, internal, external and adversarial validity.

5.8.1 Construct validity

- Checks are not credit labels. No under-collateralised lending default labels are used. Same-window checks are allowance and transfer local statistics; the holdout label is future new approvals.
- Holdout construct. Future new approvals are still endorsement events. They are out of the score window, but they are not an independent credit or trading construct.
- Own-construct agreement is partly mechanical. EndorseRank is a global smoothing of spender in-approve degree, and AWP of decayed transfer in-strength; each method’s agreement with its own family is expected by construction, which is why the within-method contrasts of Table 4.4 are reported and why the null result on them is treated as informative.

5.8.2 Internal validity

- Shared observation window. All methods and proxies are computed over 1 December 2025 to 31 May 2026. Market regimes specific to that window—volatility, incentive programmes, protocol changes—may inflate or deflate alignment. Robustness checks on damping and top-token subgraphs reduce sensitivity to hyperparameters and token concentration but do not vary the calendar window.
- Cohort selection. The matched cohort requires non-zero connectivity in both reputation subgraphs. Results may not extend to addresses absent from allowance and transfer subgraphs; the sample-definition sensitivity of Table 4.10 shows how strongly the coefficients respond when that requirement is relaxed.
- Edge-weight measurement. Token-decimal normalisation, unit conversion and permit-versus-approve coverage can affect w_{uv} . Measurement error in edge weights propagates to $\mathbf{r}$ and to rank correlations ^[23].
- Timing environment. Runtimes are wall-clock measurements on one machine; the standard deviations in Table 4.1 bound within-session noise but not between-machine variation, which

is why ratios rather than seconds are the comparative summary.

5.8.3 External validity

- Single chain and window. Evidence is limited to Arbitrum One over six months. Generalisation to other rollups, Layer-1 chains or lending-centred venues requires cross-protocol validation; DeFi venues differ in protocol design ^[8] and in market structure ^[1].
- Pool scaling versus alignment scaling. The expanded pool supports a node-count statement about runtime only (Section 4.2.1); same-window alignment remains anchored to the matched cohort, and the holdout to spenders. Efficiency gains should not be assumed to preserve the same τ profile at arbitrary population sizes.

5.8.4 Adversarial validity

- Sybil and wash trading. Adversarial allowance or transfer patterns may manipulate rankings ^[11]. Sybil-adjusted proxies partially address but do not eliminate this risk. The cost model of Section 5.6 is analytical; it is not an empirical red-team evaluation.
- Strategic approval behaviour. Once EndorseRank is deployed, rational actors may grant allowances strategically to farm rank. Observational τ estimated before deployment may overstate robustness to strategic behaviour.

5.9 Limitations

Results reflect one six-month window on Arbitrum One for a matched wallet cohort and a spender holdout cohort whose sizes and graph dimensions are given in Chapter 4. EndorseRank’s demonstrated strengths are allowance-construct alignment, rank distinctness from AWP, computational efficiency attributable to sparsity, and holdout alignment with future new approvers; its demonstrated weaknesses are weaker transfer alignment and weaker longitudinal-stability alignment than AWP, and an allowance lead over its own transfer alignment that did not reach significance.

Token-decimal normalisation, unit conversion and permit-versus-approve coverage may affect edge weights. The expanded-pool experiment tests node count, not edge growth, because the second-tier extraction did not complete. Extended baselines (alternative PageRank variants and lending-oriented graphs) are archived for future comparison (Appendix 8.4) and excluded from the main narrative. Unsecured lending default labels are out of scope. Regulatory and ethical deployment constraints (Section 5.7) limit how scores should be used even when alignment is favourable.

These limitations bound the claims: EndorseRank is a credible, efficient, allowance-specific social reputation method on the studied cohort and window; it is not a universal credit score, a Sybil-proof identity system or a replacement for protocol-level collateralisation^[9]. Chapter 6 restates what was established, what it adds to knowledge, and the work that would be needed to relax these bounds.

6 Conclusion and Future Work

Chapter 5 interpreted the results and judged the objectives met. This chapter closes the thesis: it answers each research question in one paragraph, states the original contribution to knowledge without repeating the contribution list of Chapter 1, and sets out the work that would relax the limits recorded in Section 5.9. Numbers are not restated; the tables of Chapter 4 remain the single point of reference.

6.1 Answers to the research questions

The thesis formulated EndorseRank, PageRank on a latest-allowance endorsement graph, and evaluated it against Adaptive Weighted PageRank (AWP), the transfer-graph baseline of Do, Do and Nguyen^[3], on one matched Arbitrum One wallet cohort, one spender holdout cohort and one expanded address pool over December 2025 to May 2026.

RQ1 asked whether EndorseRank and AWP produce meaningfully different wallet orderings on an identical sample. They do. The inter-method rank agreement lies strictly between unrelated and identical orderings with an interval that excludes both extremes (Table 4.3), and the rank scatter shows substantial pairwise disagreement. Hypothesis H1 is supported: authorisation graphs and payment graphs are not interchangeable reputation substrates.

RQ2 asked which method aligns more strongly with contemporaneous allowance checks and with contemporaneous transfer checks. The answer is construct-specific. AWP aligns more strongly with transfer proxies and EndorseRank with allowance proxies, and the two between-method paired differences exclude zero in opposite directions (Table 4.4). H2 is supported in that form. The within-method differences—EndorseRank’s allowance alignment against its own transfer alignment—did not reach significance, so the thesis does not claim that EndorseRank favours one signal over the

other inside its own ranking.

RQ3 asked whether EndorseRank is computationally more efficient than AWP on the same matched sample. It is, on wall-clock time, memory and iterations, and the advantage is attributable to the endorsement graph having about one ninth of the transfer graph's edges under a single timing protocol (Table 4.1). H3 is supported. The node-count sensitivity on the expanded pool confirms the ratio across staged node counts but, because the second-tier extraction did not complete, does not test edge growth (Table 4.2).

RQ4 asked whether scores frozen at t1 predict new owner–spender approvals at t2 on inbound spenders more strongly than AWP and more informatively than the same-window allowance check. They do (Table 4.11): EndorseRank's interval excludes zero and lies above AWP's, whose interval includes zero, and the labels post-date every scored event. RQ4 replaced the trading-success comparison of the research proposal, which was withdrawn as circular (Section 2.8); the holdout tests temporal predictive validity of the endorsement construct, not credit-risk validity.

Claims C1–C4 (Section 4.8) describe an efficiency–alignment trade-off. EndorseRank has lower PageRank cost, allowance-construct alignment and out-of-window evidence on future approvals. AWP has stronger transfer and longitudinal-stability alignment. Which to deploy depends on whether the operator's question is about authorisation or about flow.

6.2 Original contribution to knowledge

The contribution list of Section 1.9 named what the thesis set out to add. In the terms of the literature reviewed in Chapter 2, the following is established.

Before this work, the graph-reputation literature on public ledgers ranked wallets on transfers ^{[3],[4]}, on transaction-network risk propagation ^[12] or on explicit ratings and stake ^{[25],[26]}, and it validated those rankings against statistics drawn from the same input data. The ERC-20 allowance—the one on-chain act that records a deliberate delegation of spending authority ^[18]—had not been used as a reputation edge, and no same-cohort comparison with confidence intervals, difference tests and

a temporal holdout existed for any pair of on-chain reputation methods. This thesis supplies that edge construct and that comparison. PageRank is used unchanged ^[15]. The knowledge claim is that the choice of edge determines what an on-chain reputation score measures, that this can be shown with quantified uncertainty on one cohort, and that the allowance edge yields a score which is cheaper to refresh, distinct from the transfer-based score, aligned with its own construct and predictive of later endorsement, while weaker than the transfer-based score on flow and stability. The construct-validity framing ^{[22],[33]} that makes this a testable statement, and the open pipeline that makes it a reproducible one, are part of the contribution, because the auditability of decentralised credit-related scores is itself a recognised concern ^[2].

The research is deliberately bounded. It does not show that on-chain reputation can substitute for collateral ^{[1],[9]}, that EndorseRank is Sybil-proof (Section 5.6) or that its results hold beyond Arbitrum One and the studied window.

6.3 Future work

Each item below is tied to a limitation recorded in Chapter 5.

- Completing the second-tier extraction. The expanded pool tested node count only. Extracting approval and transfer logs for the supplemental addresses would allow runtime to be measured as the edge population grows and would turn the bounded node-count statement of RQ3 into a scaling result.
- Cross-protocol and cross-chain validation. Replication on other rollups and Layer-1 networks would test whether the allowance–transfer trade-off and the holdout pattern generalise beyond Arbitrum One; the heterogeneity of DeFi venues ^{[1],[8]} makes this the main external-validity gap.
- Longer windows and a credit-linked holdout. A longer observation window would test stability of the coefficients across market regimes, and a holdout label tied to repayment or liquidation, where lending-protocol coverage of the spender cohort permits, would test whether

the endorsement construct carries any credit-risk information; the present holdout does not.

- Adversarial evaluation. The Sybil cost model is analytical. Empirical red-teaming—wash approvals, coordinated spender farms, strategic revocation—would quantify realised attack cost-effectiveness under live fee markets and detection heuristics ^[11], and would test whether the Sybil-adjusted stability proxies remain informative under active adversaries.
- Hybrid and richer authorisation semantics. Combining endorsement-graph and transfer-graph features, or adding protocol-specific delegation logs to the edge definition, may improve coverage; such designs should report construct-specific checks and time-split evidence, not a single aggregate metric, and should preserve the auditability that distinguishes graph propagation from opaque learned scores ^{[2],[14]}.
- Deployment studies. Integration with risk oracles and governance tooling would test whether EndorseRank’s runtime profile sustains production refresh cadences, and controlled fee-tier or monitoring-cadence experiments would replace the adoption hypotheses of Section 5.4.4 with evidence. Collateral reduction remains a hypothesis, not a claim of this research.

6.4 Closing reflection

On-chain reputation does not remove the need for collateral in adversarial, pseudonymous markets ^[9]. It can make behavioural structure readable, and this research shows that the ranking depends on which edges are used. Transfers record economic flow; allowances record delegated authority. PageRank can rank either graph, but the same ranking cannot serve both constructs. EndorseRank uses the authorisation graph and reports its costs and benefits, including a time-split check of later approvals and two null within-method contrasts.

A protocol or researcher who needs a fast, auditable score of who is trusted to spend has, on Arbitrum One data of the kind studied here, a usable instrument in EndorseRank. One who needs to find transfer hubs should still use AWP. Predicting trading profit from social graphs alone is outside what either method can answer.

7 References

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline that produced the numerics in Chapter 4: data sources, processing stages, default parameters, proxy formulas, and a glossary. All quantities cited here also appear in the main-text tables and figures; no external files are required to interpret the claims.

8.1.1 Pipeline stages

The evaluation proceeds in two stages on Arbitrum One over the observation window 2025-12-01 to 2026-05-31 (UTC, inclusive).

1. Reputation graphs and scores: Extract ERC-20 Approval and Transfer logs in monthly batches (per-query dry-run and a 150 GiB abort cap). The matched cohort is wallets with non-zero approval or transfer connectivity ($n = 5,521$). Build the latest-allowance endorsement graph and the time-decayed transfer graph, then compute EndorseRank and AWP scores by power iteration (damping $d = 0.85$, tolerance 10^{-8} , at most 300 iterations).
2. Checks, alignment, and benchmarks: Compute transfer and allowance checks defined in Chapter 3, report Spearman ρ and Kendall τ against each score vector with 95% paired-bootstrap intervals (400 wallet resamples, seed 42) and the six pre-specified $\Delta\tau$ contrasts, time PageRank solves on pre-built edge lists under one protocol (one untimed warm-up, five timed repeats, per-session memoisation), and run sensitivity sweeps over damping, top tokens, and sample definition.

The tables of Chapter 4 were generated by `scripts/run_dissertation_eval.py--real--robustness--benchmark-repeats5` followed by `scripts/export_latex_results.py` on

12 September 2026; the machine-readable summary is `data/processed/eval_summary.json`. Offline synthetic checks use the same scoring and alignment logic on fixture graphs (no cloud access) to validate implementation before real extraction. Real extraction requires Google Cloud credentials and a billing project; scans target only the public Arbitrum One log dataset.

8.1.2 BigQuery extraction design

ERC-20 Approval and Transfer events are identified by their standard topic-0 hashes and restricted to logs whose owner/spender or from/to address lies in the evaluation wallet list, so scan cost scales with the cohort rather than the full chain. Monthly batching keeps each query under the operational byte cap. Representative SQL patterns appear in Appendix 8.2.

Extraction records (bytes processed, row counts, timestamps) are retained in the study’s acquisition log for auditability; this research reports the resulting cohort sizes and edge counts rather than those operational logs.

A second-tier extraction was planned for the 93,559 supplemental addresses of the expanded wallet pool ($N = 99,080$; Section 3.12.1). The dry-run estimate for the six monthly approval and transfer queries over that address set exceeded 1.3 TiB, and the download stage failed repeatedly on the storage-backed result path; the extraction was therefore abandoned within the study period. The supplemental addresses consequently enter the expanded-pool experiments as nodes only, with no approval or transfer edges of their own, which is why the full-pool row of Table 4.2 reproduces the matched-cohort graph. Completing that extraction is listed as future work in Chapter 6.

8.1.3 Scaling tables

Table 8.1 reports in-cohort runtime scaling for $n \leq 5,521$ under the timing protocol of Section 3.11; its last row is the same measurement as Table 4.1. Unlike the expanded-pool stages, every in-cohort stage is a subset of wallets whose edges were extracted, so $|V|$ and $|E|$ grow with n here.

Table 8.1: In-cohort runtime scaling (SHA256 seed benchmark-scale-v1; $n \leq 5,521$ matched GMX cohort). Here both $|V|$ and $|E|$ grow with n because every stage is a subset of wallets whose edges were extracted. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
1,000	0.203	1.874	$9.2\times$	1,302	2,584	6,463	23,058
2,000	0.312	3.995	$12.8\times$	2,362	5,147	11,849	47,031
4,000	0.802	6.869	$8.6\times$	4,645	10,557	22,591	97,105
5,521	1.426	11.253	$7.9\times$	6,442	14,727	30,791	137,087

8.1.4 Credit-delegation extract (deferred)

Aave V3 BorrowAllowanceDelegated events were not extracted in this study. Any exploratory ranking that depends on credit-delegation edges is therefore undefined on the present sample and is reported as non-interpretable ($\tau = \text{---}$) in the archived Aave-oriented matrix (Appendix 8.4). Extending the study to delegation-based reputation would require a dedicated extract of those events and a fresh alignment evaluation; it is left to future work.

8.1.5 Supplementary alignment summary table

Table 8.2 lists family-mean τ for all six proxy families computed by the pipeline, including the liquidation, inverse-risk and GMX trading-success families that were withdrawn from the main narrative together with the GF-PR reference (Section 2.8). The transfer, allowance and Sybil-stability rows are those of Table 4.3.

Table 8.2: Mean Kendall τ by proxy family: EndorseRank and AWP ($n = 5,521$).

Family	EndorseRank	AWP	Higher
Transfer	0.333	0.534	AWP
Allowance	0.355	-0.030	EndorseRank
Liq.	0.093	0.113	AWP
Inv.risk	0.035	0.107	AWP
Sybil	0.233	0.286	AWP
GMX	0.096	0.179	AWP

8.1.6 Parameter summary

Table 8.3: Default parameters used in the primary evaluation.

Parameter	Value	Role
PageRank damping d	0.85	Default teleport mixture
Convergence tolerance ε	10^{-8}	Power-iteration stop
Max iterations	300	Power-iteration cap
Logistic k	0.01	AWP decay slope
Logistic t_0	180 days	AWP decay midpoint
Observation window	2025-12-01 – 2026-05-31	UTC inclusive
Bootstrap resamples / seed	400 / 42	Paired wallet bootstrap, percentile 95% CI
Timed repeats / warm-up	5 / 1	Runtime protocol
In-cohort subsample seed	benchmark-scale-v1	Deterministic SHA256 seed
Expanded pool subsample seed	benchmark-tier2-v1	Deterministic SHA256 seed

8.1.7 Proxy operational definitions (reference)

The following formulas match Chapter 3.

- Transfer: $\text{in_degree}(w) = |\{u : \exists u \rightarrow w\}|$; $\text{in_value}(w) = \sum_{e:u \rightarrow w} \text{value}(e)$.
- Allowance: $\text{in_approve_degree}(w) = |\{o : \exists o \rightarrow w\}|$; $\text{in_approve_value}(w) = \sum_o w_a(o, w)$ with w as spender.
- Sybil-adjusted stability: $\text{inbound_counterparty_ratio}(w) = |\{u : u \rightarrow w\}| / |T^{\text{in}}(w)|$ (self-transfers excluded); tenure in days; active months.

8.1.8 Glossary

Allowance ERC-20 authorization for a spender to transfer tokens on behalf of an owner.

Alignment Rank agreement between reputation scores and a validation proxy (ρ, τ) .

Construct overlap Shared data source between a ranking method and a proxy, inflating alignment.

Endorsement graph Directed graph of latest allowances (EndorseRank input).

Matched cohort Primary sample of $n = 5,521$ wallets with non-zero endorsement- and transfer-graph connectivity.

Expanded wallet pool Larger address set ($N = 99,080$) used for node-count sensitivity of runtime; supplemental addresses have no extracted edges.

Paired bootstrap Resampling wallets with replacement using one shared index matrix for every coefficient, so that differences between coefficients can be tested on the same resamples.

PageRank Spectral ranking by stationary distribution of a damped random walk.

Proxy Observable wallet metric used for domain-intuition validation.

Teleport Uniform jump in PageRank with probability $1 - d$.

8.2 SQL Templates, Configuration, and Extended Notes

This appendix collects representative SQL patterns, configuration keys, and extended methodological notes that support reproduction without cluttering the main chapters.

8.2.1 Approval log extraction pattern

ERC-20 Approval events are identified by topic 0 equal to the keccak256 hash of Approval(address,address,uint256). A simplified monthly query pattern is:

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS owner_topic,
```

```

    topics[SAFE_OFFSET(2)] AS spender_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
AND (
    LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
    ↪ UNNEST(@wallet_list)
    OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
    ↪ UNNEST(@wallet_list)
);

```

Wallet filtering via @wallet_list keeps scans proportional to the cohort rather than the full chain. Monthly batching respects per-query byte caps documented in the study’s acquisition record.

8.2.2 Transfer log extraction pattern

Transfer events use topic 0 for Transfer(address,address,uint256):

```

SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS from_topic,
    topics[SAFE_OFFSET(2)] AS to_topic,
    data AS value_data

```

```

FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
      AND block_timestamp <  TIMESTAMP('2026-01-01')
      AND topics[SAFE_OFFSET(0)] =
      ↪ '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
      AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
        ↪ UNNEST(@wallet_list)
      );

```

Post-processing applies token decimals, builds time-decayed edge weights, and restricts nodes to the evaluation sample before PageRank.

8.2.3 Configuration keys

The pipeline’s parameters are collected in a single versioned configuration file. Salient keys include:

- observation window start and end timestamps;
- PageRank damping, tolerance, and maximum iterations;
- logistic decay k and t_0 ;
- benchmark repetition count and subsample seeds;
- output locations for intermediate artifacts and LaTeX fragments.

Changing damping or decay parameters requires re-running the full evaluation—including robustness sweeps and table export—and regenerating figures from the same artifacts, so that reported numerics remain consistent with the pipeline’s machine-readable summary.

8.2.4 Complexity notes

Let n be the number of nodes and $m = |E|$ the number of edges. Building latest-allowance edges is $O(R_a \log R_a)$ if R_a approval rows are sorted by timestamp, or $O(R_a)$ expected time with a hash map keyed by (token, owner, spender). Building transfer edges is $O(R_t)$ over transfer rows. PageRank is $O(m + mI)$ for I iterations, the first term being the assembly of the sparse operator. Iteration counts and edge counts for both methods are given in Table 4.1; EndorseRank needs roughly a third of AWP’s iterations and $m_{\text{approve}} \ll m_{\text{transfer}}$, so both factors favour EndorseRank.

Memory is dominated by edge lists and score vectors. The peak-memory ratio in Table 4.1 is of the same order as the edge-count ratio.

8.2.5 Interpretation checklist for readers

When reading Chapter 4 tables, the following checklist reduces common misreadings:

1. Identify whether a row is a social method (EndorseRank or AWP).
2. Match each family to its intended construct (allowance vs. transfer).
3. Prefer relative comparisons on the same cohort over absolute τ thresholds.
4. Read runtime tables as PageRank solve time on pre-built edge lists, not end-to-end ETL time, and expect identical numbers wherever the same graph is tabulated under the same settings (Section 3.11).
5. Read every τ with its interval, and read differences between coefficients from Table 4.4 rather than by subtracting point estimates.
6. Treat the sample-definition sweep (Table 4.10) as a sensitivity analysis whose absolute values are not comparable with the matched-cohort tables.

8.2.6 Archived extended baselines

Three-method (with the withdrawn GF-PR reference), seven-method and six-Aave alignment matrices are archived alongside the evaluation pipeline, together with corresponding CSV matrices

for spreadsheet inspection. They are excluded from the main narrative because several variants require additional event types (e.g., credit delegation) or introduce constructs outside the primary EndorseRank–AWP question. Researchers extending this work may re-enable exports via a dedicated export utility with documented presets.

8.2.7 Limitations reprise

Appendix material does not expand the empirical claims beyond Chapter 4. In particular, SQL templates do not imply that other chains were queried, and archived baselines do not constitute peer comparisons in this research. The authoritative numerics remain those in the pipeline’s machine-readable summary artifact for the matched cohort ($n = 5,521$) and expanded wallet pool ($N = 99,080$).

8.3 Elementary Properties of PageRank and Rank Correlation

This appendix records elementary properties used implicitly in Chapters 2–4. The material is standard ^{[15],[20],[21],[23]} and is included for self-contained reading.

8.3.1 Existence and uniqueness of PageRank

Let P be a row-stochastic matrix on n nodes, possibly after dangling-node repair as in Equation (2.1). For $d \in (0, 1)$, the Google matrix

$$G_{\text{PR}} = dP^\top + \frac{1-d}{n}\mathbf{1}\mathbf{1}^\top$$

is positive and column-stochastic (up to the transpose convention in Equation (2.2)). By the Perron–Frobenius theorem for positive matrices, G_{PR} has a unique eigenvalue of modulus one with a strictly positive eigenvector π normalized so that $\sum_i \pi_i = 1$. Hence PageRank scores exist and are unique for any directed graph once damping and dangling-node rules are fixed ^[23].

This uniqueness underpins comparisons between EndorseRank and AWP: given fixed parameters,

each method induces a single score vector on a given edge set. Differences between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ are therefore attributable to edge semantics and weights, not to solver multiplicity.

8.3.2 Continuity in edge weights

PageRank is a continuous function of the entries of P for fixed $d \in (0, 1)$. Small perturbations of allowance or transfer weights therefore induce small perturbations of scores in the $\|\cdot\|_1$ norm, with Lipschitz constants depending on d and n [23]. Continuity justifies robustness checks that vary damping or restrict to top tokens: if alignment were knife-edge fragile, small hyperparameter changes would scramble family-level τ . Empirically, EndorseRank’s allowance family mean is unchanged to three decimals across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.8), consistent with a stable ranking geometry.

8.3.3 Sparsity and per-iteration cost

A sparse matrix–vector product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations for m nonzero edges, and assembling the sparse operator from an edge list is likewise $O(m)$. Power iteration therefore costs $O(m + mI)$ for I iterations. On the matched cohort the transfer graph has roughly 9.3 times as many edges as the endorsement graph, so even at identical I the transfer method would be about an order of magnitude more expensive; the observed iteration counts widen the gap, and the measured wall-time ratio (Table 4.1) is of the same order as the edge ratio.

8.3.4 Properties of Spearman’s ρ

Spearman’s ρ is Pearson’s correlation applied to ranks. It is invariant to strictly increasing transformations of either variable and lies in $[-1, 1]$. Perfect monotone agreement yields $\rho = 1$; perfect reverse agreement yields $\rho = -1$. Ties require mid-rank adjustments; the evaluation pipeline uses standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association is more appropriate than Pearson correlation on raw values; the AWP baseline paper validates its ranking with the same rank statistics^[3].

8.3.5 Properties of Kendall's τ

Kendall's τ equals the normalized difference between concordant and discordant pairwise orderings. It admits a probabilistic reading: for two random distinct wallets, τ is the probability their relative order agrees under both rankings, minus the probability it disagrees ^[21]. This reading is useful when communicating results to practitioners: a τ in the mid-0.3 range, as for EndorseRank on the allowance family (Table 4.3), means a clear but imperfect tendency for the two orderings to agree.

For large n , τ and ρ are often similar in sign and magnitude but not identical. This research reports both, following the AWP baseline^[3], and uses family-mean τ with a paired-bootstrap interval ^[34] as the primary summary.

8.3.6 Construct validity note

Cronbach and Meehl^[22] distinguish criterion-related validity from construct validity. In the absence of a gold-standard default label, this research pursues a construct-oriented strategy: each method is tested against checks that operationalize its intended meaning (allowance receipt for EndorseRank; inbound transfer activity for AWP) and against a temporal holdout of later new approvals. High within-construct τ supports interpretation. Low cross-construct τ is not automatically failure; it may indicate discriminant validity.

8.3.7 Where each primary quantity is reported

To keep every number in one place, the primary quantities are not transcribed here; the following index gives the table in which each is established.

Quantity	Reported in
Matched-cohort size, $ V $, $ E $, runtime, SD, memory, iterations	Table 4.1
In-cohort scaling stages	Table 8.1
Expanded-pool stages ($N = 99,080$)	Table 4.2
Family-mean τ with 95% CI; inter-method τ	Table 4.3
Paired $\Delta\tau$ contrasts	Table 4.4
Per-proxy ρ , τ , CI	Tables 4.5, 4.6, 4.7
Damping, top-token, sample-definition sweeps	Tables 4.8, 4.9, 4.10
Holdout spender cohort ($n = 1,335$)	Table 4.11
Observation window	2025-12-01 – 2026-05-31 (Table 8.3)

All table values are rendered by the exporter from the pipeline’s machine-readable evaluation summary; none is transcribed by hand.

8.4 Archived Extended Baselines

During exploratory analysis, additional PageRank variants were scored on the same matched cohort and proxy families. They are archived here for completeness and future work. They are *not* part of the primary EndorseRank–AWP research questions and should not be read as peer claims in the main narrative.

8.4.1 Seven-method alignment matrix

Exploratory seven-method matrices remain in the evaluation archive (LP-PR, CW-AWP, LF-PR, and RiskProp-style variants). They are not drawn here. EndorseRank remains the allowance-family leader among social methods; AWP (and closely related transfer variants) lead on transfer.

Interpretation guidelines:

- Methods built on transfer-like edges cluster near AWP on transfer and Sybil families.

- Methods with very small edge counts can produce unstable or degenerate rankings; their run-times may be low simply because m is tiny.
- RiskProp-style rows that mirror AWP numerically indicate shared transfer structure, not independent discovery.

8.4.2 Six-method Aave-oriented matrix

An Aave-oriented preset remains in the evaluation archive. Several lending-event graphs are sparse on Arbitrum within the study window; delegation-based rows may be degenerate when BorrowAl-`lowanceDelegated` events were not extracted (Appendix 8.1).

The Aave-oriented preset remains in the evaluation archive and is not drawn here.

These matrices motivate future work on credit-delegation edges and cross-protocol lending labels, but they do not alter Claims C1–C4.

8.4.3 Why the main narrative uses two methods

Focusing this research on EndorseRank and AWP serves two goals:

1. Clarity of contribution: The novel construct is allowance-based endorsement. Comparing it to the established transfer baseline^[3] isolates the research question.
2. Avoiding baseline sprawl: Additional variants that share transfer edges with AWP add rows without adding independent constructs.
3. Avoiding circular references: The outcome-native GF-PR reference used in the research proposal is built from the same GMX events as the trading-success proxies, so any alignment between them is construct overlap, not validation (Section 2.8); its three-method matrix is archived alongside the seven-method and six-Aave presets.

Researchers who need broader baselines can regenerate the archive tables from the open evaluation pipeline using the seven-method and six-Aave presets.

8.4.4 Reproducing archive tables

Archive tables are regenerated by the evaluation pipeline’s table export with seven-method and six-Aave presets. CSV copies are recorded alongside the LaTeX fragments for spreadsheet inspection.

8.4.5 Notes on archived method identifiers

The following identifiers appear in archive tables and CSV exports. They document exploratory implementations in the evaluation codebase.

LP-PR A transfer-graph variant with alternative loss-penalized or liquidity-oriented weighting explored during development. Empirically it tracks AWP closely on transfer families, indicating shared payment-flow structure.

CW-AWP A counterparty-weighted AWP variant that down-weights repeated transfers between the same pair. It slightly reduces transfer τ relative to AWP while remaining in the same construct family.

LF-PR A low-frequency or lightly filtered graph with very small edge counts and near-zero runtimes; rankings are not comparable to full AWP and are retained only to document a failed sparse baseline.

RiskProp A risk-propagation style ranking on transfer-like edges inspired by Lin et al.^[12]. On this cohort it numerically mirrors AWP on several families, suggesting that the implemented risk signal did not diverge strongly from transfer centrality under the study’s parameterization.

LiqCall-PR / Borrow-PR / Repay-PR Aave-oriented wallet-to-wallet graphs built from liquidation calls, borrows, and repays when those events are present. On Arbitrum in the study window, edge counts are small (tens to hundreds), so alignment is weak and runtimes are negligible.

Delegation-PR Intended to use credit-delegation allowances; with zero delegation edges extracted, scores are degenerate and τ is not interpretable.

These notes exist so that future researchers can decide which variants deserve full re-implementation. Archive rows are not peer-reviewed baselines.

8.4.6 Decision log: what stayed in the main text

The decision to keep only EndorseRank and AWP in Chapters 4–5 was made after inspecting the seven-method and six-Aave matrices. Criteria were:

1. Construct independence: A baseline must implement a distinct edge semantics, not a minor reweighting of transfers.
2. Data adequacy: Edge counts must be large enough for stable PageRank on the matched cohort.
3. Narrative focus: This research’s contribution is allowance-based endorsement, not a survey of all possible PageRank variants.

Under these criteria, EndorseRank (allowance) and AWP (transfer) form a minimal sufficient set. Archive tables remain available for readers who want the seven-method and six-Aave matrices without mixing them into the primary claims.

8.5 Claim-to-Evidence Map

This appendix maps each hypothesis, research question, and empirical claim to the specific evidence in Chapter 4. It is intended as a navigation aid for examiners and replicators; numerical values are not repeated here.

8.5.1 H1 / RQ1 — Rank divergence

- Statement: EndorseRank and AWP yield distinct wallet orderings on the same cohort.

- Evidence: inter-method Kendall τ with 95% CI (Table 4.3, last row; Section 4.5); rank scatter in Figure 4.4; score distributions in Figure 4.3.
- Interpretation: related but not interchangeable rankings; O1 criterion met (Table 5.1).

8.5.2 H2 / RQ2 — Construct-specific alignment

- Statement: the two methods differ on contemporaneous allowance versus transfer checks.
- Evidence: family means with CI (Table 4.3); paired $\Delta\tau$ contrasts ii and iii (Table 4.4); per-proxy detail (Tables 4.5 and 4.6); heatmap Figure 4.2.
- Qualification: within-method contrasts i and vi in Table 4.4 include zero; the claim is between-method only.

8.5.3 H3 / RQ3 — Computational efficiency

- Statement: EndorseRank is faster than AWP at equal n , attributable to edge count.
- Evidence: Table 4.1 (runtime, SD, memory, iterations, $|V|$, $|E|$); Table 8.1; Table 4.2; Figure 4.1.
- Qualification: expanded-pool stages add nodes without edges (Section 4.2.1); the claim does not cover edge growth.

8.5.4 RQ4 — Temporal holdout

- Statement: after scores are frozen at t_1 , EndorseRank predicts t_2 new owner–spender approvals on inbound spenders more strongly than AWP.
- Evidence: Table 4.11 (Section 4.7).
- Qualification: the label is the endorsement construct in the future tense, not a credit or trading outcome; RQ4 replaced the proposal’s trading-success comparison (Section 2.8).

8.5.5 Claims C1–C4

- C1: runtime evidence as in H3/RQ3.

- C2: intended-construct checks—allowance for EndorseRank, transfer for AWP—in Table 4.3.
- C3: construct-specific winners as in H2/RQ2, tested in Table 4.4.
- C4: efficiency–alignment trade-off combining C1, C3 and the spender holdout (Table 4.11); robustness in Tables 4.8–4.10 and Figure 4.5.

8.5.6 Explicit non-claims

This research does *not* claim:

- that AWP is obsolete;
- that EndorseRank’s allowance alignment exceeds its own transfer alignment;
- that runtime scales favourably as the edge population grows beyond the cohort graph;
- that results generalise beyond Arbitrum One and the study window without replication;
- that on-chain scores replace collateral or legal identity systems.

8.5.7 Artifact checklist for replication

1. Merged wallet-ranking table — scores for EndorseRank and AWP.
2. Machine-readable holdout summary for the spender cohort.
3. Machine-readable evaluation summary — authoritative same-window numerics, per-run timings, bootstrap protocol.
4. Exported LaTeX table fragments (results/tables/).
5. Exported vector figures (Figures/generated/).
6. Optional archived extended-baseline matrices.

Replicators can regenerate tables and figures by running the evaluation pipeline on real data with robustness sweeps, exporting the tables, and regenerating figures from the same artifacts (Appendix 8.1.1).

8.6 Ethics Clearance Certificate

8.6.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate Reference: [Unisa SoC/CAES ethics clearance ref: TBD] Date of issue: [TBD] (Scanned certificate to be inserted here.)
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